

**Artificial Intelligence-Based Decision
Support System for Identification of Paddy
Crop Diseases and Rice Production
Analysis with Land Fertilization Prediction**

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Artificial Intelligence-Based Decision Support System for Identification of Paddy Crop Diseases and Rice Production Analysis with Land Fertilization Prediction

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of the requirements for the degree*

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by

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This thesis is dedicated with the utmost appreciation and gratitude to my beloved family, friends, and esteemed guide. Their unwavering support, encouragement, and expert guidance have been indispensable throughout my academic journey. I am deeply indebted to their belief in me, their unwavering presence, and their invaluable contributions, all of which have played a pivotal role in shaping my achievements and making this research possible.

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Abstract

Over 50% of the world's populace depends on paddy as their primary dietary source. Unfortunately, like numerous other plant species, paddy is vulnerable to a range of maladies that may reduce the quantity and quality of its yield by as much as 40%. While it is important for farmers to visually identify and diagnose these diseases to prevent crop loss, this task is often impossible due to the sheer size of many farmlands. Traditional methods allow for the extraction of insightful information that may be used to solve the current issue. To solve this problem a novel multimodal crop decision support system is proposed using the multi-head attention-based multi-input multi-output model. The proposed system consists of a paddy crop disease recognition model to detect and prevent paddy crop diseases using transfer learning, a cost estimation of paddy crop harvesting model to measure the cost associated with the crop production, and a prediction of land fertilization model using an ensemble machine learning that enables to predict the level of land fertility before the cultivation of paddy crop along with forecasting horizon. The paddy crop disease detection and recognition model accomplished a remarkable mean accuracy of 96%, having a 95% recall value and an average precision of 96%. This shows that the model exhibited excellent consistency and accuracy in paddy crop disease identification and diagnosis. The model's average precision of 94% further suggests that it was able to strike an excellent balance between precision and accuracy. Regarding the models for predicting harvesting costs and land fertilization, the average R2 score of the regression model for harvesting cost prediction was 0.95, indicating that the model was highly reliable and precise in predicting harvesting costs. Meanwhile, the land fertilization prediction model had an average accuracy of 87%, indicating that it was able to provide farmers with reliable recommendations for fertilization, taking into account multiple factors such as soil quality and weather patterns. These results indicate the efficiency and utility of the introduced decision support system for enhancing crop management and improving yield and profitability for farmers.

Keywords: Precision agriculture, Multi-modal decision support systems, Paddy crop diseases, Harvesting cost prediction, Land fertilization prediction, Multi-head attention.

Contents

List of Abbreviations	xi
List of Figures	xiii
List of Tables	xv
1 Introduction	1
1.0.1 Motivation	3
1.1 Organization of thesis	6
2 Literature Survey	7
2.1 Detection and classification of rice diseases: An automated approach using textual features [6]	8
2.2 Rice blast disease recognition using a deep convolutional neural network [7]	8
2.3 Monitoring and controlling rice diseases using Image processing [8]	9
2.4 Rice leaf disease detection using machine learning techniques [9]	9
2.5 Detection and classification of rice plant diseases [4]	9
2.6 Rice plant disease classification using color features: a machine learning paradigm [10]	10
2.7 Rice Disease Detection Using Artificial Intelligence and Machine Learning Techniques to Improve Agro-Business [11]	10
2.8 Decision support system for precision fertilization of rice based on machine learning [15]	10
2.9 Attention embedded residual for disease detection in tomato leaves [39]	11
2.10 Identification of rice diseases using deep convolutional neural networks [28]	11
2.11 Automated Image Capturing System for Deep Learning-based Tomato Plant Leaf Disease Detection and Recognition [49]	12

2.12	CNN based Leaf Disease Identification and Remedy Recommendation System [50]	12
2.13	Real-Time Detection of Apple Leaf Diseases Using Deep Learning Approach Based on Improved Convolution Neural Networks [51]	13
2.14	Identification of plant leaf diseases using a nine-layer deep convolution neural network [52]	13
2.15	A Segmentation Improved Robust PNN Model for Disease Identification in Dif- ferent Leaf [53]	14
2.16	Crop Disease Detection Using Deep Learning [54]	14
2.17	Identification of Plant Disease using Image Processing Technique [55]	15
2.18	GUI based Detection of Unhealthy Leaves using Image Processing Techniques [56]	15
2.19	Tomato Plant Leaves Disease Classification Using KNN and PNN [57]	16
2.20	Automatic Disease Symptoms Segmentation Optimized for Dissimilarity Feature Extraction in Digital Photographs of Plant Leaves [58]	16
3	Proposed Work	19
3.1	Data Preparation	23
3.1.1	Paddy Field Image Dataset	23
3.1.2	Harvesting Cost and Yield Prediction Dataset	24
3.1.3	Land fertilization Dataset	25
3.2	Preprocessing	25
4	Experimental Results	37
4.1	Experimental Setup	38
4.1.1	Hardware & Software requirements	38
4.1.2	Tools & Technologies used	38
4.2	Implementation	41
4.2.1	Experimental Evaluations (Metrics)	41
4.2.2	Hyperparameter	42
4.2.3	Experimental Setup and Model Assesment	44
4.2.4	Model Performance Evaluation	48
5	Conclusion and Future Work	53
	References	55
	Index	60

CONTENTS	ix
Publication	61
Author's Biography	63

List of Abbreviations

Adam	: Adaptive Moment Estimation
CNN	: Convolutional Neural Network
GPU	: Graphic Processing Unit
MSE	: Mean Squared Error
ReLU	: Rectified Linear Unit
LBP	: Local Binary Patterns
RoI	: Region of Interest
R2	: Coefficient of determination
HSV	: Hue, Saturation, and Value
LAB	: Lab color space or CIELAB

List of Figures

1.1	Field collected paddy crop image original data with six classes	4
3.1	Proposed Multi-Input Multi-Output Framework for Paddy Decision Support System	21
3.2	Data Preprocessing to get multi-head attention features	22
3.3	Background Removed + LBP Processing	27
3.4	HSV LAB + LBP Processing	29
3.5	Background Removed + HSV LAB Processing	29
3.6	Background Removed + HSV LAB + LBP Processing	31
3.7	HSV LAB + Background Removed Processing	31
3.8	HSV LAB + Background Removed + LBP Processing	34
4.1	Average comparison of metrics when the trained model was tested on raw images	44
4.2	Average comparison of metrics when the trained model was tested on processed images	44
4.3	Average comparison of accuracy when the trained model was tested on raw images	45
4.4	Average comparison of VGG16 and VGG19 model's accuracy on processed data	45
4.5	Average comparison of accuracy for original and processed data	45
4.6	Average comparison of precision for original and processed data	45
4.7	Average comparison of recall for original and processed data	46
4.8	Average comparison of VGG16 and VGG19 model on the basis of accuracy . . .	46
4.9	Performance comparison of VGG16 and VGG19 models on Background Removed + HSV Lab dataset	50
4.10	Performance comparison of VGG16 and VGG19 models on HSV lab Background Removed dataset	50

4.11 Performance comparison of VGG16 and VGG19 models on HSV lab Background	
Removed dataset	51
4.12 Performance comparison of VGG16 and VGG19 models on HSV lab Background	
Removed LBP dataset	51

List of Tables

3.1	Meity Paddy Crop Field Image Dataset	24
3.2	Yield and Harvesting Cost Prediction data	24
3.3	Land Fertilization Prediction Data	25
4.1	Comparison of Validation and Testing Accuracy on Different Models: Comparative Analysis with Related Research Papers	44
4.2	Comparing Results of Models on Different Datasets: Performance Evaluation across Various Experimental Settings	47
4.3	Comparison of Training and Testing R2-Score Results on Various Models: Evaluating Performance Across Different Algorithms	49

Introduction

Preface

This chapter serves as an introduction to the topic of paddy crop, covering its types, diseases, and management practices. It highlights the need for advancements in paddy crop management and introduces the problem statement. Additionally, it discusses the importance of paddy yield prediction, calculating harvesting costs, and land fertilization prediction. The chapter concludes with an overview of the organization of the thesis.

Smart agriculture is gaining more proliferation due to huge crop production for sustainability and healthy growth of the nation. As a global staple, paddy is one major survival meal consumed throughout the world, notably in Asian nations. Due to the lack of best practices and solutions for smart farming, the crop yield is reduced by nearly about 36% because of plant diseases, weeds, and pests. This significantly reduces crop yield, creating an epidemic and dire circumstance for farmers. The reason for crop loss is sporadic adherence to agricultural practices, late detection of crop diseases, and infestation by vegetation and pests. For these issues to be effectively resolved, technological intervention is essential. Several researchers have concentrated on sensor-based technology, the Internet of Things (IoT), artificial intelligence (AI), and machine learning to reduce crop loss. These algorithms provided an efficient and cost-effective solution that will improve farming and assure profitability and longevity. Moreover, a number of researchers have proposed models for disease detection and agricultural production cost estimation in an effort to reduce the number of crop losses caused by various diseases. To solve the problem of early diagnosis of plant diseases, scientists from around the world have developed disease detection and warning models based on obtained images of plant leaves and machine learning techniques. A framework for the early detection of crop maladies was proposed by researchers. The framework is trained by an SVM classifier to classify multiple types of paddy crop maladies, such as sheath blight, bacterial leaf blight, brown spot, and leaf smut, which have been identified. Further, segmenting a leaf image using the K-means clustering technique, the extracted features are used in an SVM model for classification with an accuracy of 73.33%. However, the most significant drawback of this proposed framework is that it requires more time for pre-processing and the accuracy gained by the framework using explicitly extracted features, like RGB hue features, GLCM (Grey Level Co-Occurrence Matrix) textural features, and area-shaped features, is quite low.

Through 10-fold cross-validation, paper [21] combined Bayes and SVM algorithms as a classification algorithm to determine paddy diseases, achieving a mean accuracy of 79.5%. However, this method's computation process is complex, limiting its pervasive application.

In the paper [22] they used an artificial neural network (ANN) to detect paddy plant maladies, with three output nodes for identifying leaf blast, leaf blight, and sheath blight. Their recognition accuracy was 74.21%, but there is room for improvement in their methodology.

While in the paper [23] utilized machine learning techniques to train three classifiers for measuring crop disease incidence and severity: Extremely Randomized Trees (Extra Trees), K-Nearest Neighbours (KNN), and the Linear Support Vector Classifier (Linear SVC). Although their method performed well, it lacks an end-to-end plan and heavily depends on elements that

were personally built.

In [24], image segmentation was accomplished using K-means clustering, while feature extraction was accomplished using a hybrid technique. They achieved the maximum accuracy for classification of 98.63% for identifying paddy illnesses using an SVM model with several classes. The computational process is challenging since this approach is constrained by prior experience with image processing techniques such as segmentation, filtering, and feature extraction.

1.0.1 Motivation

Deep learning (DL), a recent innovation in machine learning, in particular, a deep convolutional neural network (CNN), due to its record-breaking performance, has emerged as the preferred approach for taking care of both major and minor jobs [3]. Due to their ability to automatically extrapolate visual details and identify and classify plant illnesses using an end-to-end methodology, CNN methods have grown in favor. In the paper, [2] a powerful CNN system was developed to distinguish between the four rice leaf diseases brown spot, tungro, and leaf blast. The highest F1 score, 98.38%, through the use of the ResNet50 combined with SVM architecture. Due to the dimensions of the model, this deep CNN-based approach doesn't work well with mobile terminal development. To identify six vegetable diseases against a crowded background, a recurrent attention model was developed in [31]. It achieved an average identification accuracy of 98.26%. However, the quantity of plant species evaluated in this study is only two, tomato and cucumber, and it does not attempt to detect crop illnesses.

In the paper [27] developed a deep CNN model capable of identifying 26 illnesses and 14 crop species. On a held-out test set, the mean accuracy of the model was 99.35%. In contrast, it is challenging to deploy this deep CNN-based model in embedded devices, because all of the test photographs were taken in controlled settings with a plain background rather than in genuine field settings. Then in [28] they built a powerful CNN model using 500 original natural photos to identify 10 common illnesses of rice plants with a 95.48% accuracy rate, among them are rice blast, false smut, and bacterial leaf blight. The robustness and complexity of this CNN-based model might be improved, and it is not ideal for use with mobile devices. PD2SE-Net, which was proposed by [31], is a network for diagnosing plant diseases and estimating their severity. Its core network architecture is made up of ResNet50 and shuffle units, and it achieves an overall 99%, 98%, and 91% mean accuracy for classifying paddy crop diseases as well as assessing their severity, respectively. Although this technique achieves outstanding performance, the testing materials only include photographs taken against plain backgrounds rather than in true wild field circumstances.

It is challenging to develop a strong, affordable, and cutting-edge system while taking a farmer's economic and technical issues into account. It was challenging to calibrate the newly created soil moisture sensor because the goal was to at least match the baseline performance of the current soil moisture sensor, if not outperform it. Since the sensors in the sensor node operate at a voltage substantially lower than the output of the solar cell, power conversion from the solar cell to the sensor node must be done with great caution. After the embedding system circuitry was taken care of, the training of an effective deep-learning model for classifying plant diseases with the curse of dimensionality in mind was given priority.

Due to the scale of vast rice fields, it is impractical for farmers to manually inspect each rice plant, and even if they attempted to do so, it would be prone to error and could cause damage to the plants. In addition, it is difficult to physically identify and diagnose plant problems due to the need to observe various parameters, such as environmental conditions and environments. To address this issue, researchers are investigating researchers and farmers may benefit from the potential of artificial intelligence (AI) and machine learning (ML) in the early detection of diseases in rice plants. This technological approach can significantly aid in the early identification of problems and provide agriculture with a more efficient and effective means of support [4]. Figure 1.1 depicts the Field images of the paddy crop dataset with six disease classes.

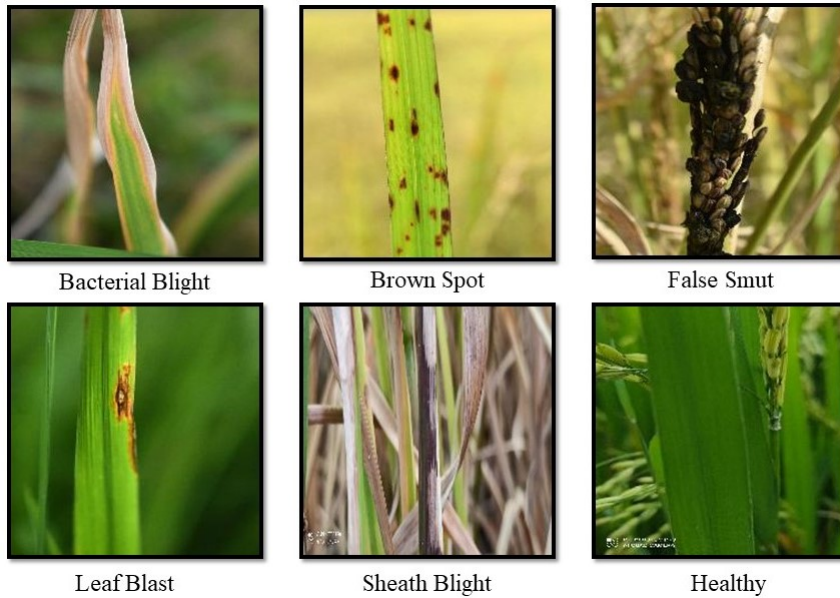


Figure 1.1: Field collected paddy crop image original data with six classes

Creating a system to aid paddy farmers in the early identification of rice disease serves an obvious purpose, as it can lead to increased production and quality while reducing costs, ben-

efiting both farmers and consumers. In the agricultural technology research field, the primary focus is on improving production and quality, and rice plants require particular attention due to their unique characteristics. Therefore, research in this area is critical to prevent and detect rice plant diseases, which can have significant impacts on the food supply for millions of people. The goal of this work is to develop a system that automatically detects abnormal behavior, categorizes it, and diagnoses it using cutting-edge artificial intelligence (AI) and deep learning (DL) approaches to paddy diseases with high accuracy, without human intervention. Additionally, the aim of this study is to provide new approaches that can achieve higher diagnostic accuracy than existing techniques with similar datasets, also this study is not limited to disease detection but we further propose methods through which farmers can estimate the harvesting cost of the field considering various parameters that impact rice production. Harvesting cost prediction is another critical aspect of paddy crop management, as it allows farmers to plan and budget effectively. Harvesting costs can vary significantly depending on factors such as farm size, machinery type, and labor costs. Predictive models can be developed using machine learning algorithms to estimate the costs associated with harvesting, allowing farmers to make informed decisions about when to harvest. We identified one more problem farmers usually face before starting the production season of any field, the problem lies with whether the field supports the crop productivity or not. Accurate land fertilization estimation is also crucial for optimal paddy crop management. Paddy farming requires careful land management, and the right balance of fertilizers is crucial for achieving optimal yields. We proposed predictive models to help farmers and other stakeholders determine the optimal amounts and types of fertilizers to apply to their land, taking into account factors such as soil quality, weather patterns, and crop rotation. This can lead to more efficient use of fertilizers, better crop yields, and cost savings. That means inspection regarding fertilization needs to be done before starting the actual harvesting. So we have addressed these issues as part of this research.

In this paper, a multi-modal decision support system for paddy crop management is proposed that combines automated disease recognition, harvesting cost prediction, and land fertilization prediction using machine learning techniques. Our system employs techniques for deep learning, such as convolutional neural networks, to develop an automated disease recognition model that can accurately identify and classify paddy crop diseases. Additionally, we use machine learning algorithms, such as regression and multi-head attention, to develop predictive models for harvesting costs and land fertilization. Our system provides a comprehensive solution for paddy crop management, allowing farmers to make informed decisions and optimize their yields.

1.1 Organization of thesis

The following sections of this work are organized as follows: Paddy crop disease and other machine-learning applications are highlighted in Section II's discussion of the literature on machine learning and deep learning in agriculture. The proposed technique is thoroughly explained in Section III. Section IV contains a presentation and discussion of all the experimental findings. The study is concluded in Section V, which also suggests possible areas for further research.

Literature Survey

Preface

This chapter includes some of the previous works that is done in crop disease detection including different crops as well as various disease identification techniques. Finally a table to conclude all the literature works presented.

Image processing is a crucial component of machine learning (ML) algorithms that help classify images into different categories by identifying common features. Preprocessing, feature extraction, and classification are the three phases of ML algorithms, with the classification phase relying on either supervised or unsupervised methods. Images are input into deep learning (DL) algorithms, which extract features and categorize them. DL algorithms have grown in popularity for image processing. In order to tackle research challenges, ML and DL algorithms are both often utilized in many different domains.

2.1 Detection and classification of rice diseases: An automated approach using textural features [6]

In the paper [6], the authors suggest using the Support Vector Machines (SVM) classifier to categorize three types of rice crop diseases - brown spots, false smuts, and bacterial leaf blight. They recommend using the Scale-Invariant feature transform (SIFT) and Bag of Word (BoW) methods to extract features. They also propose applying K-means clustering and Brute-Force (BF) matcher in addition to SVM for classification. Their dataset consists of 400 images from different sources such as the Rice Research Institute (RRI), the Rice Knowledge Bank (RKB), and the American Psychopathological Society (APS). They reported average accuracy of 94.16 %, recall of 91.60%, and precision of 90.9%. However, given that multiclass classification is being considered, their dataset is rather tiny, and SVM is a classifier that is susceptible to overfitting.

2.2 Rice blast disease recognition using a deep convolutional neural network [7]

The authors of the publication [7] propose using a deep convolutional neural network (CNN) to find cases of rice blast illness. They utilized a publicly available dataset containing 5,812 images that were equally divided between rice plants that are diseased and uninfected. Their method uses an SVM for classification and a CNN to extract features, yielding an average accuracy of 95.83% for binary classification.

2.3 Monitoring and controlling rice diseases using Image processing [8]

Similarly, in a paper [8], the authors suggest utilizing image processing techniques to monitor and manage rice disease, targeting four specific types: brown patches, blast, and a bacterial blight on rice, as well as rice sheath. They recommend using both conventional classifiers, including k-Nearest Neighbour (k-NN) and Minimum Distance Classifier (MDC), as well as designed features based on form and color. They only have 115 photos of these disorders in their dataset, which is divided into 30% for testing and 70% for training. 89.23% accuracy for MDC and 87.02% accuracy for k-NN were reported by the authors. It's important to note that their dataset is comparatively small for multiclass classification and that overfitting is not a concern in their work.

2.4 Rice leaf disease detection using machine learning techniques [9]

The authors of [9] recommend using machine learning techniques to distinguish between the three primary types of diseases that affect rice leaves: bacterial leaf blight, leaf smut, and brown spot. They utilize a dataset of 120 images with an equal distribution of images for each disease, which was provided in [4]. Several traditional classifiers are proposed, including Naive Bayes, J48 Decision Tree, Decision Tree, Logistic Regression, and K-nearest Neighbour. The authors report that J48 Decision Tree achieves the highest accuracy of 97.9%. However, the small dataset size might have an impact on this outcome.

2.5 Detection and classification of rice plant diseases [4]

The authors of [4] advise segmenting the contaminated area of the leaf using k-means clustering and extracting features based on texture, shape, and color. They show an average accuracy of 93.33% on training data and 73.33% on testing data while using Support Vector Machines (SVM).

2.6 Rice plant disease classification using color features: a machine learning paradigm [10]

In [10], the authors use color features for the classification of rice plant disease. They analyze 14 color spaces and extract four color features from each channel, resulting in a total of 172 features. They use a dataset consisting of 619 images with four classes: rice blast, bacterial leaf blight, healthy leaves, and sheath blight. The authors then use seven classifiers to test their method, including Logistic Regression, Random Forest, Decision Tree, Naïve Bayes, K-nearest neighbor, SVM, and discriminant classifier. With an average accuracy of 94.65%, they report the greatest SVM accuracy.

2.7 Rice Disease Detection Using Artificial Intelligence and Machine Learning Techniques to Improve Agro-Business [11]

In [11], there is a comprehensive review of several AI and ML methods used for detecting paddy diseases. This is important because rice is a globally significant crop. Meanwhile, [12] presents a swifter regional CNN (Faster R-CNN) that quickly detects rice leaf diseases in real-time by utilizing the regional proposal network (RPN). They tested their method using both existing datasets and their own dataset, which included a total of 2400 images and achieved high accuracy rates for different types of rice diseases as well as healthy leaves. Specifically, they achieved accuracy rates of 99.17% for hispa, 98.85% for brown spots, 98.09% for rice blast, and 99.25% for healthy leaves.

2.8 Decision support system for precision fertilization of rice based on machine learning [15]

A study in the paper [15] developed a decision support system predicting using machine learning methods and providing optimal fertilization recommendations. Another study in the paper [16] suggested a deep learning-based method for recognizing and categorizing rice illnesses. Multi-head attention has been used in several studies for crop yield prediction and optimization. For instance, a study in the paper [17] proposed a crop yield prediction model based on multi-head attention and graph convolutional networks. Similarly, a study in the paper [18] developed a crop recommendation system based on multi-head attention and gradient boosting decision tree. Regression analysis is a commonly used technique for harvesting cost

prediction in agriculture. A study in the paper [19] proposed a decision support system that combines regression analysis and machine learning algorithms for crop yield and profitability prediction. Another study in the paper [20] developed a regression-based model. CNN methods have gained popularity due to their ability to automatically extract image features and perform end-to-end identification and classification of plant diseases. [30] developed a deep CNN-based system to identify four types of paddy leaf diseases, achieving an optimum F1 score of 98.38% using the ResNet50 plus SVM architecture. However, this deep CNN model is not suitable for deployment in mobile applications due to its large size.

2.9 Attention embedded residual for disease detection in tomato leaves [39]

In the context of vegetable diseases, A proposed RA-CNN mode in [39] employs progressive learning and region proposal to recognize six vegetable diseases in cluttered background conditions, achieving a 98.26% recognition accuracy on average. Instead of disease detection, this approach, however, only focuses on disease identification. Additionally, just two plant species (cucumber and tomato) are tested. ([27]) introduced a deep CNN model with a high accuracy of 99.35% on a held-out test set that can identify 14 crop species and 26 illnesses. However, this deep CNN-based model is difficult to implement in embedded devices, and the examined photographs were taken in controlled settings with a plain background as opposed to in actual field settings.

2.10 Identification of rice diseases using deep convolutional neural networks [28]

In the paper ([28]) 500 natural photos were used to train a deep CNN model to recognize ten prevalent illnesses of rice plants with an accuracy of 95.48%. This deep CNN-based technique, however, is not appropriate for deployment on mobile devices, comparable to earlier models, and there is room to increase its robustness and complexity. A plant disease diagnostic and severity estimation network, known as PD2SE-Net, was proposed by [31]. Its main network architecture comprises ResNet50 and shuffle units. For classifying diseases, identifying plant species, and determining severity, they attained high overall accuracies of 0.99, 0.98, and 0.91, respectively. However, rather than using real-world field circumstances, the experimental data used in this study only consists of photographs taken against plain backgrounds.

2.11 Automated Image Capturing System for Deep Learning-based Tomato Plant Leaf Disease Detection and Recognition [49]

Innovative technology, such as smart farming systems that make use of relevant infrastructure, has the potential to improve the quality and quantity of agricultural production in the country, particularly tomato production, in the long run. Because tomato plant farming takes into account a variety of aspects such as the atmosphere, the soil, and the amount of sunshine received, the presence of diseases cannot be completely avoided. The current advancement in computer system innovation, made possible by deep learning, has paved the path for the detection of tomato leaf disease captured by a camera. This paper resulted in the development of a unique approach for the efficient detection of illness in tomato plants. Several tomato plants were imaged using a motor-controlled image capturing box to detect and characterise leaf diseases on all four sides of each tomato plant. The experiment was conducted on a specific tomato variety known as Diamante Max, which was utilised as the test subject. Phoma Rot, Leaf Miner, and Target Spot were the illnesses targeted by the system, which was meant to identify them. It is collected both damaged and healthy plant leaves for inclusion in the dataset. Then you'll need to train a deep convolution neural network to recognise three different disorders. The method used a Convolution Neural Network to determine which of the tomato illnesses was present on the tomato plants that were being monitored by humans. With an accuracy of 95.75 percent, the F-RCNN trained anomaly detection model outperformed the Transfer Learning illness recognition model. A real-world implementation of the automated image capturing system resulted in a 91.67 percent accuracy rate in the recognition of tomato plant leaf diseases.

2.12 CNN based Leaf Disease Identification and Remedy Recommendation System [50]

The agricultural field has a significant impact on our daily lives. In our country, agriculture is the most significant sector of the economy. Farmers have a tough time identifying the leaf disease, which results in their producing less. Agriculture experts can provide a better answer, even though videos and photographs of leaves provide a clearer view of the situation. As a result, the problem associated with crop disease can be resolved. It is important to highlight that if the crop's yield is reduced as a result of illness, the crop has a significant risk of failing to provide

adequate nourishment. Because of advancements and developments in technology, equipment that are intelligent enough to recognise and diagnose plant illnesses have become available. Recognize diseases and treat them as soon as possible in order to reduce the detrimental influence on harvest. The authors of this work concentrate on the detection of plant diseases using image processing techniques. This research makes use of an open dataset that contains 5000 photos of healthy and diseased plant leaves, and it employs semi-supervised approaches to distinguish between crop types and detect disease in four different classes.

2.13 Real-Time Detection of Apple Leaf Diseases Using Deep Learning Approach Based on Improved Convolution Neural Networks [51]

This paper discusses five different varieties of apple leaf disease: aria leaf spot, brown spot, mosaic, grey spot, and rust. Aeria leaf spot is the most common type of apple leaf disease. This is a problem with the apple. The authors of this research applied deep learning approaches to increase the performance of convolution neural networks (CNNs) for the identification of apple leaf diseases. A new apple leaf disease detection model that uses deep-CNNs is developed in this paper using the Apple Leaf Disease dataset (ALDD), which contains complex images and laboratory images, with the remainder constructed using data augmentation and image annotation technologies. The model is developed by using Rainbow concatenation and Goo-gle Net Inception structure to detect apple leaf disease. The proposed INAR-model is trained on the testing dataset, which contains 26,377 photos of apple leaf disease. The model is then utilised to detect five prevalent apple leaf illnesses. The testing findings reveal that the INAR- SSD model achieves a detection performance of 78.80 percent, while also achieving a high detection speed of 23.13 frames per second. As a result of the findings, the innovative INAR-SSD model is demonstrated to be a high-performance solution for the early diagnosis of apple leaf illnesses, capable of performing real-time detection of these diseases with higher accuracy and faster detection speed than earlier techniques.

2.14 Identification of plant leaf diseases using a nine-layer deep convolution neural network [52]

In this paper, they have demonstrate how to identify plant leaf disease using deep learning techniques in a convolution neural network (CNN). The Convolutional neural network model is

trained utilising more than 39 different classes of open datasets of plant leaf illnesses, as well as background photos and is capable of learning new things when new data is added. That include six different types of data augmentation methods, which are used for gamma correction, image flipping, principal component analysis (PCA), colour augmentation, rotation, noise injection, and scaling [4] among other things. The fact that data augmentation is being used is being seen by all. This has the potential to improve the model's performance. The model was trained with multiple training ranges of epochs, batch sizes, and dropouts, as well as varied dropout rates. After that, CNN is compared to transfer learning methodologies, with the suggested model achieving a superior outcome.. When working with the validation data, Despite the fact that the proposed simulation model achieves 96.46% classification accuracy , Compared to transfer learning techniques, the accuracy of the CNN is significantly higher.

2.15 A Segmentation Improved Robust PNN Model for Disease Identification in Different Leaf [53]

This paper offers Agricultural Images of vegetables, fruits, crops, and flowers, as well as images of leaf disease. Identification of diseases related with specific agricultural commodity types. Symptoms of these illnesses are particular to the product component, which might be any of the following: root, seed, or leaf. This is useful in providing illness identification from a remote lab, which is beneficial. The job is separated into two stages in this section. The segmentation model for leaf images is defined in the first stage, and it is used to investigate the features of leaf images. Once the features have been found, the work is applied to the PNN classifier in order to determine the presence or absence of disease. The job is about to identify the health and sickness that has been infected based on the identification of the featured region. The approach is applied to leaf photographs from the web that have been randomly selected for different plants.

2.16 Crop Disease Detection Using Deep Learning [54]

Crop diseases have seen a significant increase in recent years as a result of extreme climate changes and a lack of immunity in crops. Farmer's financial losses result as a result of large-scale crop destruction, decreased cultivation, and other factors. Identifying and treating diseases has become a significant difficulty as a result of the rapid increase in the number of diseases and the lack of basic knowledge among farmers. The texture and visual similarity amongst the

leaves are characteristics that aid in the identification of the illness. As a result, computer vision combined with deep learning offers a solution to this problem. Using a publicly available collection of photographs of healthy and damaged crop leaves, this research presents a deep learning-based model that can be trained. By categorising images of leaves into diseased or uninfected categories based on the pattern of defect, the model accomplishes its goal.

2.17 Identification of Plant Disease using Image Processing Technique [55]

Agriculture has evolved into something much more than a means of providing food for an ever-increasing population. It is critical in countries where agriculture provides employment to more than seventieth of the population, as is the case in many Asian countries. As a result, it provides a diverse spectrum of persons. The first and most important thing to consider is that disease has reduced the amount of high-quality crop produced. The early diagnosis of leaf disease may be able to prevent agricultural losses. In this case, the goal is to design a software system solution that can automatically detect and classify disease. There are several phases involved in illness detection, including picture acquisition, pre-Processing, segmentation, extraction, and classification, among others. The photographs of the leaves are used for the detection of plant diseases. Because of this, image processing techniques are being used to identify and classify diseases in agriculture.

2.18 GUI based Detection of Unhealthy Leaves using Image Processing Techniques [56]

Increasing agricultural productivity has a positive impact on the Indian economy. This study contributes to the identification of diseased leaf utilising image processing techniques in order to achieve an efficient and smart farming system, with the goal of achieving an efficient and smart farming system. Ladies finger plant leaves are selected and analysed in this manner in order to detect the early stages of several illnesses such as yellow mosaic vein, leaf spot, powdery mildew, and so on. In order to determine if a leaf is healthy or diseased, photos of the leaves are taken, analysed, segmented, and classed according to their attributes. Because of practical limits in terms of climatic conditions and other terrain regions, noisy picture data sets are also generated and taken into account in the analysis. For segmentation, K-Means clustering is employed, and for classification, SVM and ANN are used, respectively. PCA is

used in this work to minimise the number of features. It is demonstrated in these results that the average precision of detection in SVM and ANN is 85 percent and 97 percent, respectively. They are seen to be 92 percent and 98 percent, respectively, when there is no noise present. Achieving total automation in agricultural industries is made possible thanks to this approach.

2.19 Tomato Plant Leaves Disease Classification Using KNN and PNN [57]

Plant diseases are a significant factor in agricultural productivity, and they have a negative impact on food security as well as farmers' profits. Plant diseases must be identified in order to be prevented from being lost by adequate feeding strategies to heal the diseases as soon as possible and to avoid a reduction in production. The authors of this research used two approaches for the identification and classification of healthy and sick tomato leaves in order to reach their conclusions. When employing the k-nearest neighbour strategy, a tomato leaf is rated as either healthy or sick in the first of these techniques. Later, in the second strategy, they identify the sick tomato leaf using a probabilistic neural network and the k-nearest neighbour approach[9] to determine whether it is healthy or unhealthy. Classification is accomplished through the use of characteristics such as GLCM, Gabor, and color. In this study, the authors used their own dataset to conduct their experiments. That consists of 600 leaves, both healthy and sick. Following the investigation, it was discovered that the fusion approach combined with a PNN classifier beats all other methods.

2.20 Automatic Disease Symptoms Segmentation Optimized for Dissimilarity Feature Extraction in Digital Photographs of Plant Leaves [58]

A critical step in the application of machine learning for the identification of plant diseases is the segmentation of unhealthy symptoms regions in images of plant leaves. It is sometimes referred to as Region of Interest (ROI) segmentation since it entails the separation of simply colour variant symptom lesions from surrounding green tissue, from which discriminant traits may then be retrieved. Investigations have revealed that a segmented ROI does not capture the vivid anatomy of a disease's symptom progression from origin to manifestation, which is necessary for the development of finer disease characterisation dissimilarity characteristics. The standard ROI segmentation method is also frequently beset by difficulties ranging from

intrinsic elements such as picture acquisition conditions to exogenous factors such as disease architecture where symptoms fade into healthy green tissue and the separation barrier becomes impalpable, among other things. Processes become more complex as a result, and the results can be inaccurate. In this study, we present an automatically extended region of interest (EROI) segmentation method that incorporates symptom progression information by extending the border region to include some healthy tissue and applying colour homogeneity thresholding to segment the region of interest. Ground truth was created using the traditional ROI segmentation method in conjunction with a reduced ROI method on a well-known Plant Village dataset, from which independent textural and colour characteristics were collected and utilised to train a linear classifier. The advantages of the proposed strategy for extracting dissimilarity features were further highlighted by a comparison of the classification outcomes. Finer characterisation traits for plant diseases may be derived through this research, which can then be used to classify them and estimate their severity, among other things.

Proposed Work

Preface

This chapter details the complete process of constructing a model for disease recognition in millet crops using deep learning and presents that model in the following chapter. This contains the intricacies of the design as well as the complete process described in depth.

In this research, we present a novel multimodal crop decision support system using a multi-head attention-based multi-input multi-output model (shown in Fig. 3.1). The proposed system consists of a paddy crop disease recognition model to detect and prevent paddy crop diseases using transfer learning, a cost estimation of paddy crop harvesting model to measure the cost associated with crop production, in addition to this a prediction of land fertilization model employing machine learning methods that allow for the prediction of the fertility of the land before the cultivation of paddy crop along with forecasting horizon. Our approach for paddy disease recognition and detection is based on an adaptation of the transformer model architecture using a multi-head attention technique for disease prediction in paddy crops.

The components include embedding layers for converting input data into continuous vectors, multi-head attention for attending to different input parts simultaneously, feed-forward networks for capturing complex patterns, and residual connections with layer normalization for information flow. We configure input and output, combining encoded data and utilizing classification layers. Then, trained the model using multiple features from different preprocessing algorithms, labeled data, optimized parameters, and validate with a separate set. After this the trained model was deployed for inference, predicting diseases using multi-head attention-based features.

Then, for crop yield estimation and early land fertilization prediction, we used multi-head attention features to extract useful features with machine learning algorithms. For paddy crop harvesting cost prediction, the proposed system uses multi-head attention to combine data on various factors such as farm size, seeds used, urea, phosphate used, and total working hours of labor, etc. to provide more precise estimates of the harvesting costs. After collecting the public paddy statistical dataset with the provided features, we started with data cleaning and feature selection stages. In the data cleaning, we eliminate inconsistencies and missing values in order to ensure the dataset's integrity.

Subsequently, feature selection techniques were utilized to identify the most pertinent features for predicting paddy crop harvesting costs and estimating yield. In this particular case, we used mutual info regressor serves as the feature selection method, leveraging mutual information as a statistical measure to quantify the interdependence between variables. By employing the mutual info regressor, the algorithm assesses the mutual information between each feature and the target variable (harvesting costs or yield), ultimately selecting the features with the highest mutual information scores. Once feature selection was completed, a diverse set of regression models, including linear regression, ridge regression, lasso regression, decision tree regression, K-nearest neighbors (KNN), and XGBoost, were employed for prediction purposes. These mod-

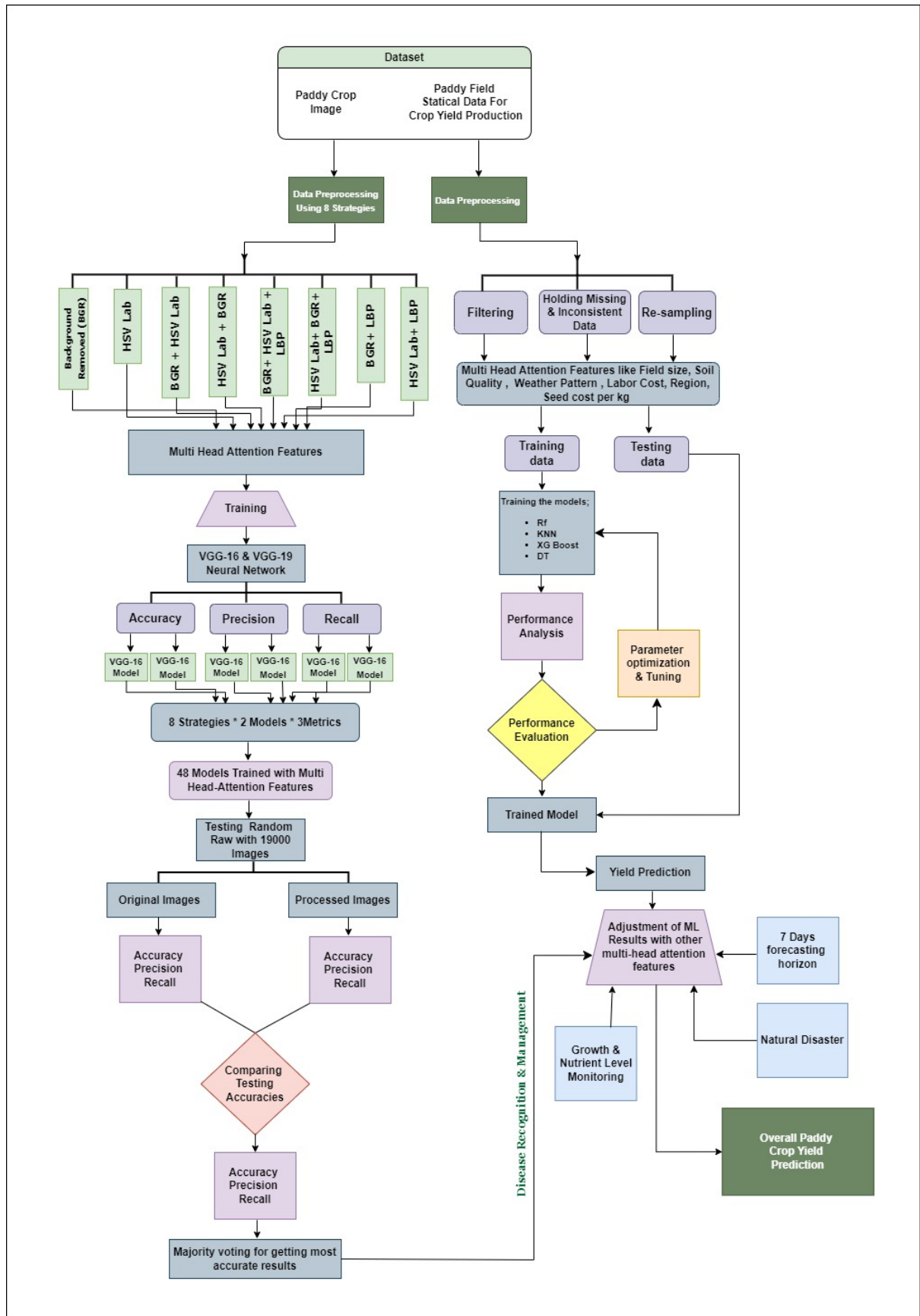


Figure 3.1: Proposed Multi-Input Multi-Output Framework for Paddy Decision Support System

els offer a range of techniques and flexibility to capture the goal variable and the input features are related in a variety of ways.

The effectiveness of each model was assessed using k-fold cross-validation. This validation technique partitions the dataset into k equally sized folds, enabling an assessment of the model's generalization ability to unseen data. The models are trained on k-1 folds and evaluated on the remaining fold, and this process was repeated k times to ensure that each fold serves as the validation set at least once. The performance of each model was ultimately assessed using the R-squared (R^2) score, which measures the proportion of variance in the target variable that can be explained by the model. A higher R^2 score indicates a better fit of the model to the data, signifying greater accuracy in predicting paddy crop harvesting costs and estimating yield. By employing this comprehensive approach, which encompasses data cleaning, feature selection, multiple regression models, k-fold cross-validation, and the use of the R^2 score as an evaluation metric, the algorithm ensures a robust and accurate analysis of the paddy crop dataset to provide reliable predictions and estimations.

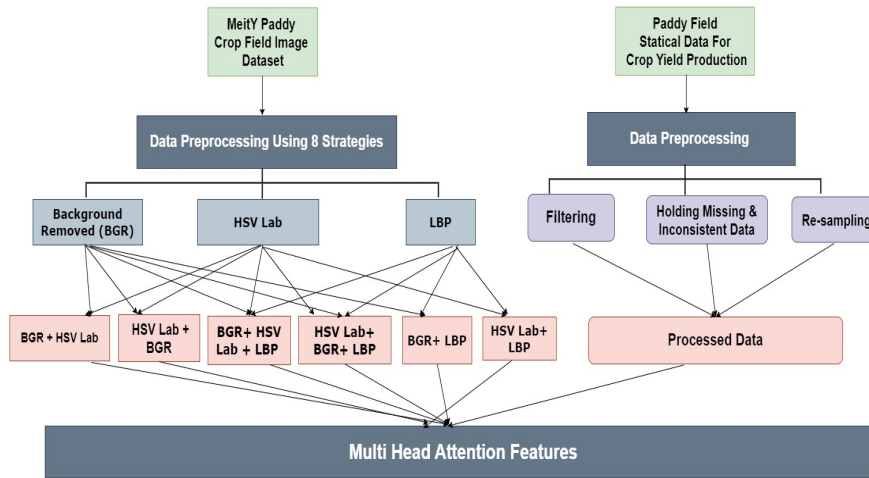


Figure 3.2: Data Preprocessing to get multi-head attention features

Similarly, for land fertilization prediction, multi-head attention is used to integrate information on different aspects such as soil quality, weather patterns, and crop rotation, to offer optimal recommendations to farmers regarding the type and quantity of fertilizers to use. Thus, different factors are being considered and provide farmers with timely and accurate information for informed decision-making, leading to improved crop yields and cost savings.

In this module, we aim to predict the values of fertilizer components Nitrogen (N), Phos-

phors (P), and Potassium (K) based on input parameters such as, crop types and crop location. To accomplish this, we first split the fertilizer dataset into training and testing datasets. The training dataset comprises 80% of the original fertilizer data to train the model for the prediction of N, P, and K values, while the remaining 20% of the datasets are used for testing the unknown (query) N, P, and K values. Subsequently, three separate random forests (RF) classifier is used. Each RF classifier consists of 50 decision trees, for predicting N, P, and K. The overall prediction for each decision tree is measured by averaging the accuracy of the individual decision trees.

The dataset containing a total of $n = 2200$ entries is divided into training and test datasets. The training set is created by selecting 80% of the data, resulting in 1,760 entries, while the remaining 20% forms the test set, consisting of 240 entries. Next, the random forest regression algorithm is employed to predict the values of N (Nitrogen), P (Phosphorus), and K (Potassium). For each of these three values, 50 estimators, or decision trees, are utilized. The N Label, P Label, and K Label refer to the individual models created for predicting the Nitrogen, Phosphorus, and Potassium values, respectively. Each label is trained using the training dataset, with the dependent variable set as N for the N Label, P for the P Label, and K for the K Label.

During the training process, each N Label, P Label, and K Label generates an output consisting of 50 decision trees, which are derived from the training dataset. These decision trees collectively form the random forest regression model for each respective label. This procedure ensures that the random forest regression model is trained and capable of predicting the values of Nitrogen, Phosphorus, and Potassium based on the provided dataset.

3.1 Data Preparation

3.1.1 Paddy Field Image Dataset

This study utilized a dataset consisting of images of six different types of paddy crop leaf diseases, including bacterial leaf blight, leaf blast, brown spot, false smut, and sheath blight, as well as healthy rice leaves. We prepared a dataset that consisted of 44,569 leaf images of paddy crops, including healthy crops and those affected by five types of diseases: (1) Bacterial Leaf Blight, (2) Brown Spot, (3) Leaf Blast, (4) False Smut, and (5) Sheath Blight. It is illustrated in Table(3.1). The dataset was split into training, testing, and validation sets, with 70% used for training, 20% for testing, and 10% for validation. First, the datasets were highly imbalanced, which could have affected the model's ability to accurately predict all classes. Secondly, all the entries in the CSV file corresponding to the original dataset were maintained

Class	Count
Healthy	4575
Sheath Blight	2859
Leaf Blast	1893
False Smut	13821
Bacterial Blast	7862
Brown Spot	13559

Table 3.1: Meity Paddy Crop Field Image Dataset

and read consecutively for a label, which could have further contributed to a falsely high result. To address these issues, several solutions were proposed. First, all the imbalanced processed datasets were balanced to improve the model’s performance in all classes. Then, all the entries for the original dataset were shuffled along with their corresponding labels to prevent the models from being biased toward certain classes.

3.1.2 Harvesting Cost and Yield Prediction Dataset

Now, for training the model for the prediction of the harvesting cost of paddy crops we used statistical public data for rice production from (Kaggle) and modified this data by adding certain features and removing other features as required for calculating harvesting cost. This dataset comprises 19 predictor variables and 1 response variable. The predictor variables include Id, Size, Status, Varieties, Bimas, Seed, Urea, Phosphate, Pesticide, Pseed, Purea, Pphosph, Hiredlabor, Famlabor, Wage, Goutput, Noutput, Price, and Region. During the data cleaning phase, our focus was on eliminating inconsistencies and addressing missing values to maintain the integrity of the dataset. Moving forward, we employed feature selection techniques to identify the most relevant features for predicting paddy crop harvesting costs and estimating yield.

Features	20
Shape	1026×20
Training	80%
Validation	10%
Testing	10%

Table 3.2: Yield and Harvesting Cost Prediction data

3.1.3 Land fertilization Dataset

The statistical dataset (Kaggle) for crop analysis includes input features such as the type of crop (rice, cotton, etc.), temperature in degrees Celsius, humidity as a percentage, and rainfall in millimeters. We again modified the data we required for the training. The output features consist of the ratios of Nitrogen, Phosphorous, and Potassium content in the soil. Data preparation involves cleaning the dataset by handling missing values and outliers, ensuring data consistency, and normalizing numerical features. Additionally, categorizing the crop type appropriately and scaling the data may be required. These steps are essential for analyzing and modeling the dataset to predict the nutrient ratios in the soil based on environmental factors.

Features	8
Shape	2200×8
Training	70%
Validation	20%
Testing	10%

Table 3.3: Land Fertilization Prediction Data

3.2 Preprocessing

The proposed system employs multiple preprocessing techniques for the analysis of prepared paddy crop leaf diseases using image processing. These techniques aim to enhance the quality of the captured leaf images by reducing noise and eliminating artifacts. To remove the background from the leaf images, we applied contour methods and Otsu's thresholding. Additionally, we utilized the HSV color space transformation to process RGB images of the leaves. Moreover, we employed the Local Binary Pattern (LBP) method to extract distinctive features from the preprocessed images, facilitating the classification of diseases. By employing these various preprocessing approaches, we aim to improve the accuracy and effectiveness of disease identification and analysis in paddy crop leaves.

The algorithm 1 starts by converting the input image I to grayscale (I_{gray}) and performing Canny edge detection to obtain an edge map (E). A morphological operation is then applied to enhance the edges (E_{enhanced}). Contours (C) are extracted from the enhanced edge map, and the largest contour (C_{largest}) is identified based on area calculations. A binary mask (M) is generated using the selected contour, and the mask is applied to the original image to obtain a masked image (I_{masked}). Local Binary Pattern (LBP) is computed on the grayscale image to obtain an LBP texture image (I_{lbp}). Finally, the final foreground texture image (I_{final}) is

Algorithm 1: Background Removed + LBP Processing

Input: Input image

Output: Foreground leaf image with Background Removed + LBP processing

The input and output of this processing are shown in figure(3.3)

1. Let I be the input image of size $M \times N$ with pixel values $I(x, y)$ for $0 \leq x < M$ and $0 \leq y < N$.
 2. $I_{\text{gray}} \leftarrow \text{ConvertToGrayscale}(I)$;
 3. $E \leftarrow \text{CannyEdgeDetection}(I_{\text{gray}})$;
 4. $E_{\text{enhanced}} \leftarrow \text{ApplyMorphologicalOperation}(E)$;
 5. $C \leftarrow \text{FindContours}(E_{\text{enhanced}})$;
 6. $A_{\text{largest}} \leftarrow 0$;
 foreach *contour* c *in* C **do**
 $A_{\text{contour}} \leftarrow \text{CalculateArea}(c)$;
 if $A_{\text{contour}} > A_{\text{largest}}$ **then**
 | $A_{\text{largest}} \leftarrow A_{\text{contour}}$;
 | $C_{\text{largest}} \leftarrow c$;
 7. $C_{\text{selected}} \leftarrow C_{\text{largest}}$;
 8. $M \leftarrow \text{GenerateMask}(C_{\text{selected}})$;
 9. $I_{\text{masked}}(x, y) \leftarrow \begin{cases} I(x, y), & \text{if } M(x, y) = 1 \\ 0, & \text{otherwise} \end{cases}$;
 10. $I_{\text{lbp}} \leftarrow \text{ComputeLBP}(I_{\text{gray}})$;
 11. $I_{\text{final}}(x, y) \leftarrow I_{\text{lbp}}(x, y) \cdot M(x, y)$;
-

obtained by element-wise multiplication of the LBP image and the mask. This algorithm allows for the extraction of the foreground texture while removing the background.

In the algorithm 2, we discussed how hsv lab and lbp processing. Let I be the input image of size $A \times B$ with pixel values $I(x, y)$ for $0 \leq x < A$ and $0 \leq y < B$. The input image is then converted to the HSV color space (I_{hsv}) and the LAB color space (I_{lab}). The saturation channel (S) and value channel (V) are extracted from the HSV image, and the A channel (A) and B channel (B) are extracted from the LAB image. Median filtering is applied to the saturation, value, A, and B channels to remove noise. Thresholding is performed on the filtered saturation channel (S_{filtered}) and the filtered A channel (A_{filtered}) to obtain binary images (S_{binary} and A_{binary}). The binary masks (M) are created by multiplying the binary images. The value

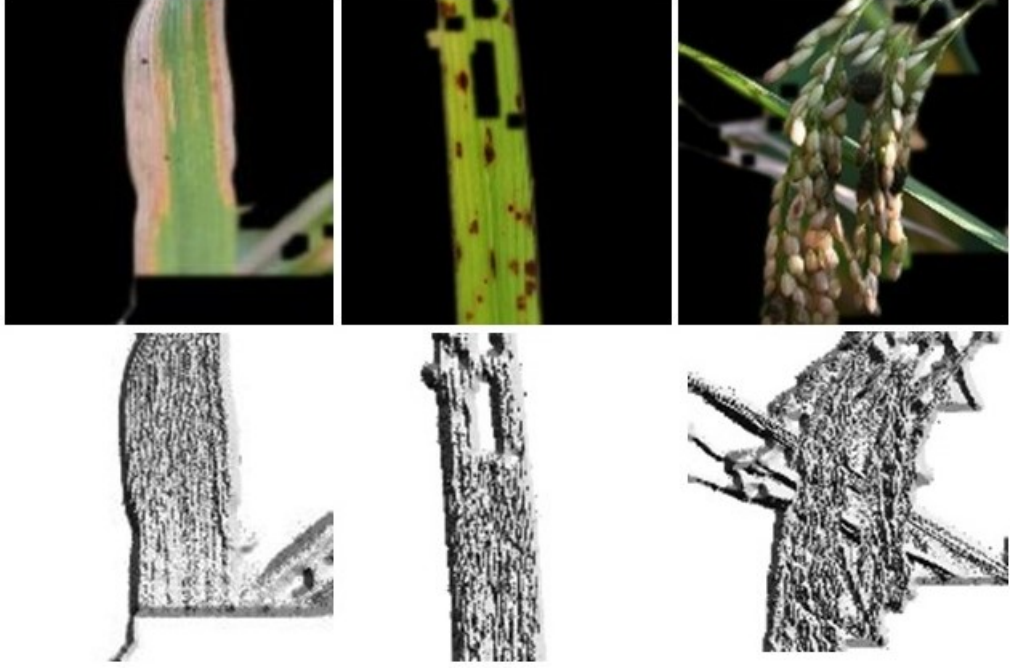


Figure 3.3: Background Removed + LBP Processing

channel is masked using M to obtain V_{masked} , and the B channel is masked using M to obtain B_{masked} . Local Binary Pattern (LBP) is computed on the masked value channel to obtain I_{lbp} , and specific patterns are identified in I_{lbp} to obtain the final output image (I_{final}). This algorithm processes the input image in HSV and LAB color spaces, performs filtering and thresholding, and uses LBP to identify patterns for the final output.

The algorithm 3 takes an input image and performs background removal and HSV LAB processing on the foreground object. It begins by converting the image to grayscale and then applies a binary thresholding operation to separate the foreground from the background. A morphological closing operation is used to fill in gaps in the foreground object and make it more connected. The resulting binary image is then used as a mask to isolate the foreground object from the original image. Finally, the isolated object is converted to the HSV color space for better color representation and analysis.

Algorithm 2: HSV LAB + LBP Processing

Input: Input image

Output: Foreground leaf image with HSV LAB + LBP processing

The input and output of this processing are shown in figure(3.4)

1. Let I be the input image of size $A \times B$ with pixel values $I(x, y)$ for $0 \leq x < A$ and $0 \leq y < B$.
 2. $I_{\text{hsv}} \leftarrow \text{ConvertToHSV}(I)$;
 3. $I_{\text{lab}} \leftarrow \text{ConvertToLAB}(I)$;
 4. $S \leftarrow \text{ExtractChannel}(I_{\text{hsv}}, \text{Saturation})$;
 5. $V \leftarrow \text{ExtractChannel}(I_{\text{hsv}}, \text{Value})$;
 6. $A \leftarrow \text{ExtractChannel}(I_{\text{lab}}, A)$;
 7. $B \leftarrow \text{ExtractChannel}(I_{\text{lab}}, B)$;
 8. $S_{\text{filtered}} \leftarrow \text{ApplyMedianFilter}(S)$;
 9. $V_{\text{filtered}} \leftarrow \text{ApplyMedianFilter}(V)$;
 10. $A_{\text{filtered}} \leftarrow \text{ApplyMedianFilter}(A)$;
 11. $B_{\text{filtered}} \leftarrow \text{ApplyMedianFilter}(B)$;
 12. $S_{\text{binary}} \leftarrow \text{Threshold}(S_{\text{filtered}})$;
 13. $A_{\text{binary}} \leftarrow \text{Threshold}(A_{\text{filtered}})$;
 14. $M(x, y) \leftarrow S_{\text{binary}}(x, y) \cdot A_{\text{binary}}(x, y)$
 15. $V_{\text{masked}}(x, y) \leftarrow \begin{cases} V_{\text{filtered}}(x, y), & \text{if } M(x, y) = 1 \\ 0, & \text{otherwise} \end{cases}$;
 16. $B_{\text{masked}}(x, y) \leftarrow \begin{cases} B_{\text{filtered}}(x, y), & \text{if } M(x, y) = 1 \\ 0, & \text{otherwise} \end{cases}$;
 17. $I_{\text{lbp}} \leftarrow \text{ComputeLBP}(V_{\text{masked}})$;
 18. $I_{\text{final}} \leftarrow \text{IdentifyPatterns}(I_{\text{lbp}})$;
-

The algorithm 4 aims to extract a foreground leaf image with background removal, HSV LAB color space conversion, and Local Binary Patterns (LBP) processing. It begins by removing the background from the original input image using a suitable method. Then, the background-removed image is converted to the HSV color space. From the HSV image, the algorithm extracts the Hue, Saturation, and Value channels. Next, LBP texture extraction is applied to the Value

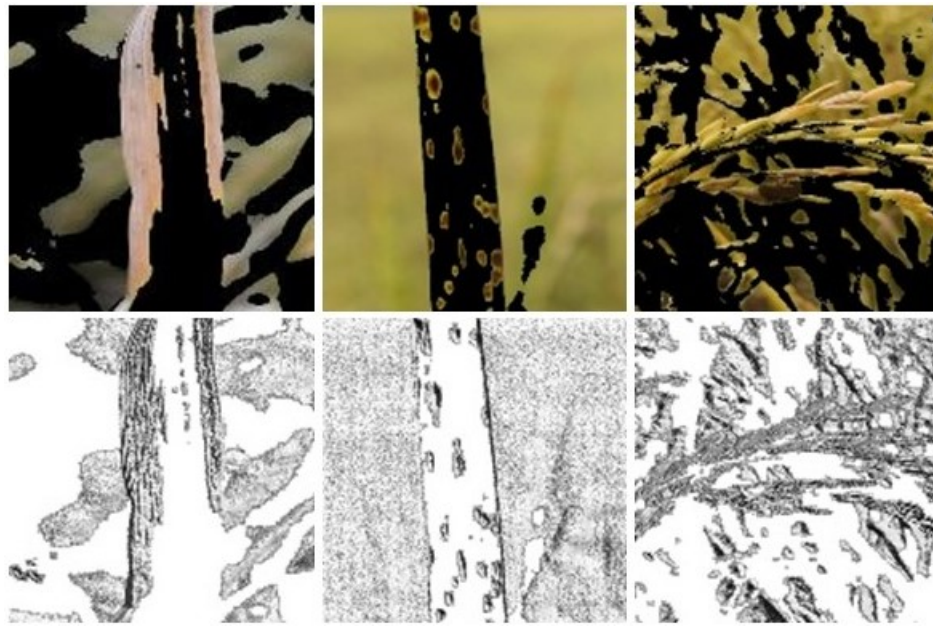


Figure 3.4: HSV LAB + LBP Processing

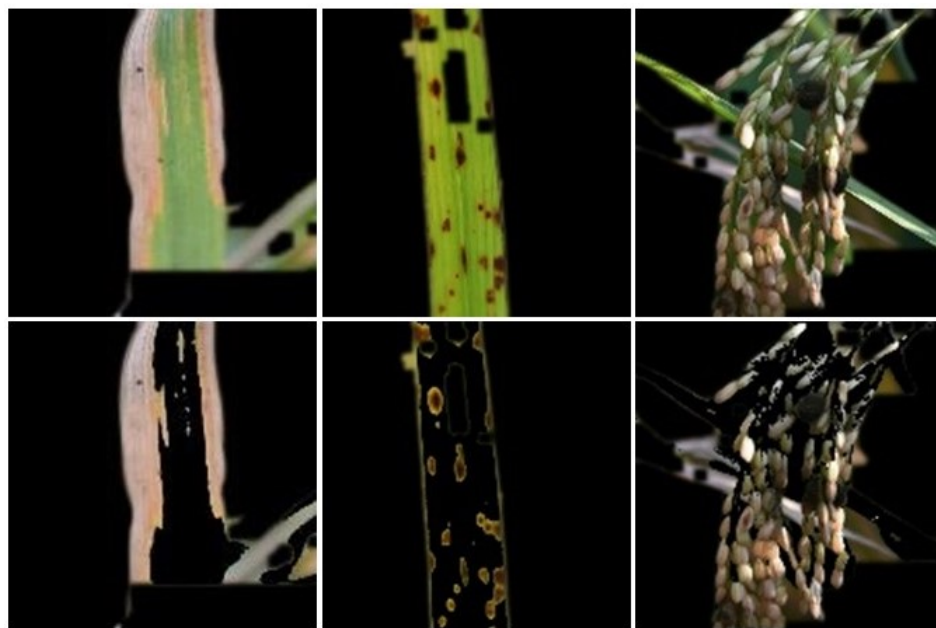


Figure 3.5: Background Removed + HSV LAB Processing

channel, resulting in an LBP texture version of the image. The algorithm selectively masks out the background-removed portion in the LBP texture image and generates a corresponding mask image. By applying this mask to the original LBP texture image, the algorithm removes the background from the texture version. The subsequent steps involve identifying minute patterns and specific disease-related patterns in the texture version, comparing the texture between

Algorithm 3: Background Removed + HSV LAB Processing

Input: Input image

Output: Foreground leaf image with Background Removed + HSV LAB processing

The input and output of this processing are shown in figure(3.5)

1. Let I be the input image of size $M \times N$ with pixel values $I(x, y)$ for $0 \leq x < M$ and $0 \leq y < N$.
2. Convert the input image to grayscale: $I_{\text{gray}}(x, y) = \frac{I(x, y)}{255}$ for $0 \leq x < M$ and $0 \leq y < N$.
3. Apply a binary thresholding operation to segment the foreground object from the background. Let T be the threshold value. The binary image B is defined as:

$$B(x, y) = \begin{cases} 1, & \text{if } I_{\text{gray}}(x, y) > T \\ 0, & \text{otherwise} \end{cases}$$

for $0 \leq x < M$ and $0 \leq y < N$.

4. Perform a morphological closing operation on the binary image B to fill in any gaps in the foreground object. This operation helps to smooth the edges and make the object more connected.
5. Let M be the resulting binary image after the closing operation. $M(x, y)$ represents the pixel value at coordinates (x, y) for $0 \leq x < M$ and $0 \leq y < N$.
6. Apply the mask M to the original image I to obtain the isolated foreground object. The masked image I_{masked} is defined as:

$$I_{\text{masked}}(x, y) = \begin{cases} I(x, y), & \text{if } M(x, y) = 1 \\ 0, & \text{otherwise} \end{cases}$$

for $0 \leq x < M$ and $0 \leq y < N$.

7. Convert the isolated foreground object I_{masked} to the HSV color space. This transformation allows for better color representation and analysis of the foreground object.
-

diseased and non-diseased portions of the image, and finally outputting the foreground texture image with the background removed.

The algorithm 5 aims to extract a foreground leaf image by applying HSV LAB color space conversion and background removal processing. The input image is initially converted from the RGB color space to the HSV color space. The algorithm then extracts the individual Hue, Saturation, and Value channels from the HSV image. A threshold is applied to the Value channel to segment the leaf from the background, creating a binary image. A morphological closing operation is performed on this binary image to generate a binary mask. By inverting

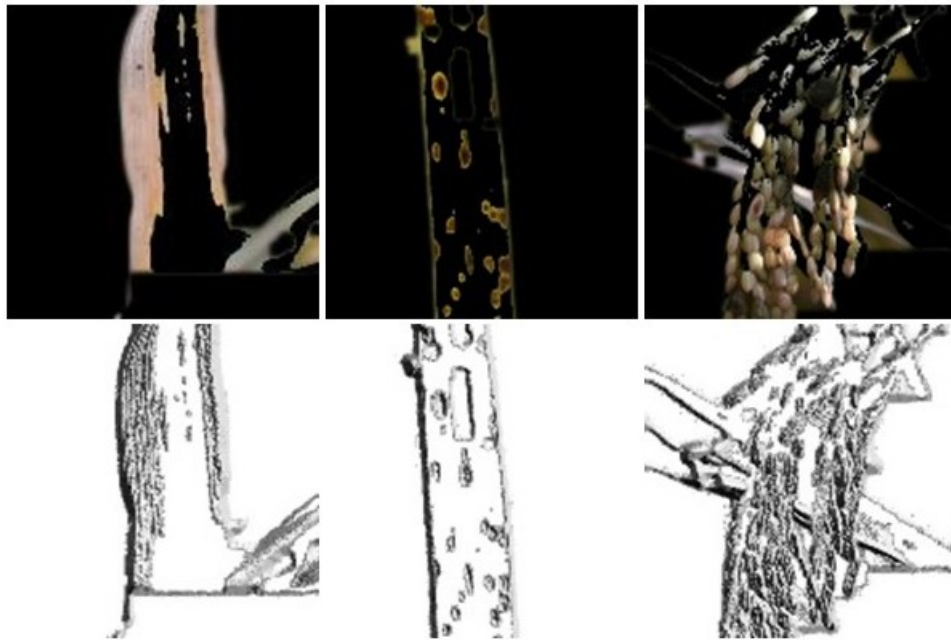


Figure 3.6: Background Removed + HSV LAB + LBP Processing

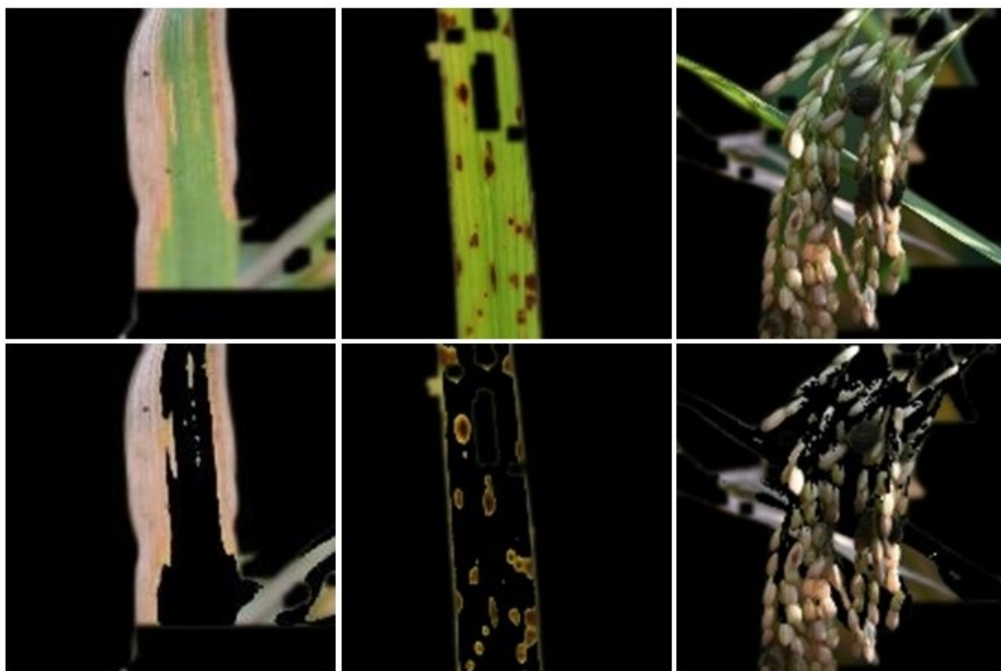


Figure 3.7: HSV LAB + Background Removed Processing

the mask, a background mask is obtained. The algorithm applies this background mask to the Hue and Saturation channels, effectively removing the background from the image. Finally, the modified image is converted back to the RGB color space, resulting in a foreground leaf image with the background removed.

Algorithm 4: Background Removed + HSV LAB + LBP Processing

Input: Input image

Output: Foreground leaf image with Background Removed + HSV LAB + LBP Processing

The input and output of this processing are shown in figure(3.6).

1. Remove the background from the original image using a suitable method. Let I be the original image of size $M \times N$ with pixel values $I(x, y)$ for $0 \leq x < M$ and $0 \leq y < N$.
2. Convert the resulting background-removed image to the HSV color space. Let I_{hsv} be the HSV version of the image.
3. Extract the Hue, Saturation, and Value channels of the image. Let H , S , and V be the extracted channels from I_{hsv} .
4. Apply LBP (Local Binary Patterns) texture extraction on the Value channel of the image. Let I_{lbp} be the LBP texture version of the Value channel.
5. Set the pixel intensity value to 1 in the texture version (I_{lbp}) of the background-removed portion of the image:

$$I_{\text{lbp}}(x, y) = \begin{cases} 1, & \text{if pixel} \in \text{BGR portion} \\ I_{\text{lbp}}(x, y), & \text{otherwise} \end{cases}$$

for $0 \leq x < M$ and $0 \leq y < N$.

6. Generate a mask (M) of the same size as the original image with the background-removed portion of the image masked out:

$$M(x, y) = \begin{cases} 0, & \text{if pixel} \in \text{BGR portion} \\ 1, & \text{otherwise} \end{cases}$$

for $0 \leq x < M$ and $0 \leq y < N$.

7. Apply the mask image (M) to the original image (I) to remove the background from the texture version (I_{lbp}):

$$I_{\text{lbp}}(x, y) = I_{\text{lbp}}(x, y) \cdot M(x, y)$$

for $0 \leq x < M$ and $0 \leq y < N$.

8. Identify any minute patterns in the texture version (I_{lbp}) of the image that may not be visible in the original colored version.
 9. Check for any specific patterns corresponding to the disease in the texture version (I_{lbp}).
 10. Compare the texture between the diseased and non-diseased portions of the image.
 11. Output the foreground texture image (I_{lbp}) with the background removed.
-

Algorithm 5: HSV LAB + Background Removed Processing

Input: Input image**Output:** Foreground leaf image with HSV LAB + Background Removed Processing

The input and output of this processing are shown in the figure(3.7)

1. Convert the input image from the RGB color space to the HSV color space:

Let I_{RGB} be the input image in RGB color space, and I_{HSV} be the image converted to HSV color space using the function $\text{ConvertToHSV}(I_{\text{RGB}})$.

2. Extract the Hue, Saturation, and Value channels from the HSV color space:

Let H , S , and V be the individual channels extracted from I_{HSV} .

3. Apply an appropriate threshold on the Value channel to segment the leaf from the background:

Let T be the threshold value. Create a binary image B by applying the threshold on the Value channel:

$$B(x, y) = \begin{cases} 1, & \text{if } V(x, y) \geq T \\ 0, & \text{otherwise} \end{cases}$$

4. Generate a binary mask from the thresholded image:

Let M be the binary mask obtained by applying a morphological closing operation to B .

5. Invert the binary mask to obtain the background mask:

Let M_{inv} be the inverted binary mask obtained by subtracting M from a matrix of ones with the same size as M .

6. Apply the background mask on the Hue and Saturation channels to remove the background from the image: Let H_{removed} and S_{removed} be the Hue and Saturation channels with the background removed, computed as:

$$H_{\text{removed}}(x, y) = H(x, y) \cdot M_{\text{inv}}(x, y)$$

$$S_{\text{removed}}(x, y) = S(x, y) \cdot M_{\text{inv}}(x, y)$$

7. Convert the image back to the RGB color space: Let I_{RGB} be the image with the background removed, obtained by converting H_{removed} , S_{removed} , and V back to the RGB color space using the function $\text{ConvertToRGB}(H_{\text{removed}}, S_{\text{removed}}, V)$.

8. Output the foreground texture image with the background removed.
-

This algorithm focuses on extracting a foreground leaf image with HSV LAB color space conversion, background removal, and Local Binary Patterns (LBP) processing. The input image is initially converted from the RGB color space to both the HSV and LAB color spaces. The LAB image is then processed to remove the background using an appropriate method, resulting in the LAB background-removed image. LBP texture extraction is applied to this background-removed

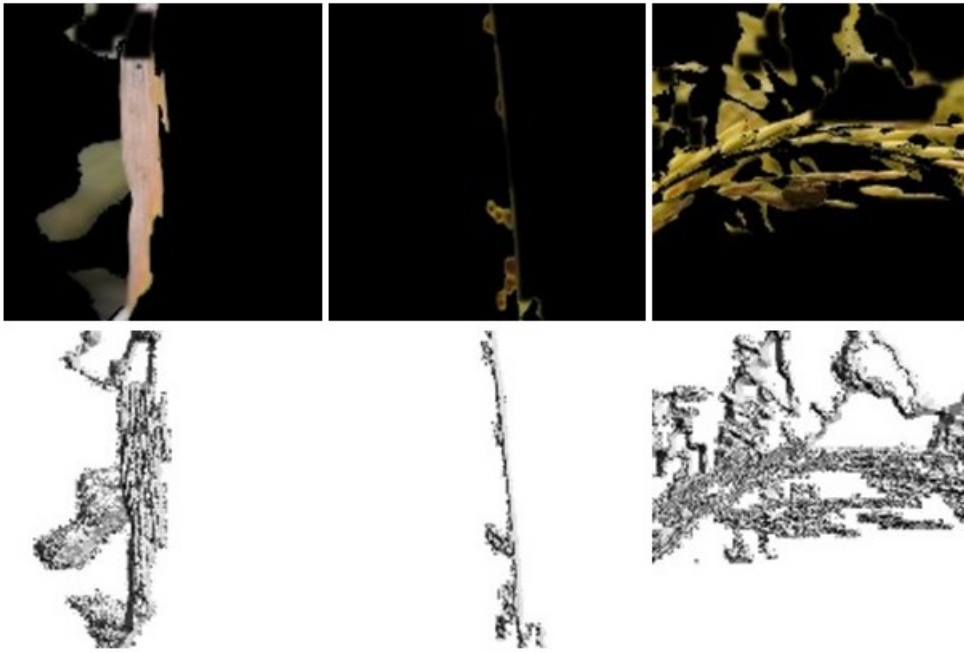


Figure 3.8: HSV LAB + Background Removed + LBP Processing

LAB image, generating an LBP texture version. The algorithm sets the pixel intensity value to 1 in the texture version for the background-removed portion. A binary mask is generated, masking out the background-removed portion in the LAB image. This mask is applied to the original LAB image, removing the background from the texture version. The algorithm then focuses on identifying minute patterns and specific disease-related patterns in the texture version of the LAB image. Finally, the algorithm outputs the foreground texture image with the background removed, incorporating LBP processing.

The algorithms are the eight different strategies used to preprocess the data and now it is ready to get trained. Initially, we started with 25,092 original images and later after these eight strategies, we have eight different datasets for testing which preprocessing suits our disease pattern recognition.

Now, moving forward with disease recognition to crop yield production. To preprocess this statistical field data for the crop yield production prediction problem, we begin with cleaning the dataset to remove duplicates and irrelevant data and fill in the missing values. Next, converting non-numerical data into numerical data and scale, select, and engineer features that are likely to have a significant impact on crop yield production. Additionally, we have included a nutrient monitoring module that predicts land fertilization, as it can have a significant impact on crop yield production.

To capture complex relationships between different features in the dataset, we used multi-

Algorithm 6: HSV LAB + Background Removed + LBP Processing

Input: Input image

Output: Foreground leaf image with HSV LAB + Background Removed + LBP Processing

The input and output of this processing are shown in figure(3.8).

1. Convert the input image from RGB color space to HSV and LAB color spaces: Let I_{RGB} be the input image in RGB color space. Convert I_{RGB} to HSV color space to obtain I_{HSV} using the function $\text{ConvertToHSV}(I_{\text{RGB}})$. Convert I_{RGB} to LAB color space to obtain I_{LAB} using the function $\text{ConvertToLAB}(I_{\text{RGB}})$.
2. Remove the background from the LAB version of the image using a suitable method: Let $I_{\text{LAB_bg_removed}}$ be the LAB version of the image with the background removed, obtained using a suitable background removal method.
3. Apply LBP texture extraction on the resulting background-removed LAB image: Let I_{LBP} be the LBP texture version of $I_{\text{LAB_bg_removed}}$ obtained using the function $\text{ApplyLBP}(I_{\text{LAB_bg_removed}})$.
4. Set the pixel intensity value to 1 in the texture version of the background-removed portion of the LAB image: Update I_{LBP} by setting the pixel intensity value to 1 in the background-removed portion:

$$I_{\text{LBP}}(x, y) = \begin{cases} 1, & \text{if pixel} \in \text{BGR portion} \\ I_{\text{LBP}} & \text{otherwise} \end{cases}$$

5. Generate a mask of the same size as the original image with the background-removed portion of the LAB image masked out: Let M be the binary mask of the same size as the original image obtained by setting the background-removed portion to 0:

$$M(x, y) = \begin{cases} 0, & \text{if pixel} \in \text{BGR portion} \\ 1, & \text{otherwise} \end{cases}$$

6. Apply the mask image to the original image to remove the background from the texture version of the LAB image: Let $I_{\text{LAB_texture_removed}}$ be the LAB image with the background removed, obtained by element-wise multiplication of I_{LAB} and M .
 7. Identify any minute patterns in the texture version of the LAB image that may not be visible in the original colored version.
 8. Check for any specific patterns corresponding to the disease in the texture version of the LAB image.
 9. Output the foreground texture image with the background removed and LBP processing applied.
-

head attention features that apply a multi-head attention mechanism to the input features, allowing the model to focus on different parts of the input at different times. With this multi-

head attention to features, the model can capture the interactions between different nutrients, environmental factors, and crop characteristics that contribute to crop yield production. This can result in more accurate predictions and a better understanding of the complex relationships that exist between the different features in the dataset. The data is now split into training, validation, and testing sets, and augmentation and dimensionality reduction techniques were applied to increase the dataset size and reduce computational complexity. Finally, normalizing the data to ensure that all features have a similar scale. These preprocessing techniques will ensure that the dataset is well-prepared for training a machine learning algorithm with multi-head attention features.

Experimental Results

Preface

This chapter focused on the implementation of all the experiments carried out as well as the experimental setup which includes hardware & software requirements, tools & technologies used. The implementation details contain the evaluation matrices and hyperparameter used.

4.1 Experimental Setup

The experimental setup contains hardware & software requirements, tools & technologies used and dataset split details.

4.1.1 Hardware & Software requirements

The experiments and related analysis processes are done on a computer with the configuration of $2\times$ CPU-4248, $4\times$ 16 GB DDR4 2933ER, $2\times$ 2TB SATA, $1\times$ SSD-1TB S4510, $3\times$ NVIDIA-RTX2080-11GB.

Throughout my research work, I have extensively relied on the Jupyter Notebook as a versatile and open-source development IDE. Its integration of live code, equations, visualizations, and narrative text has enabled me to efficiently cleanse and manipulate image data, perform numerical simulations, develop statistical models, visualize results, and implement machine learning using libraries like OpenCV and Keras. The Jupyter Notebook has proven invaluable in achieving accurate and reliable outcomes for my research in image processing.

4.1.2 Tools & Technologies used

- **Flask:** Flask, a lightweight and versatile web framework ([48]) for Python, simplifies the development of web applications. It provides a robust set of tools and libraries that enable developers to build scalable and dynamic web services and APIs. Flask's simplicity and minimalistic design make it a popular choice for web development projects. One of the key advantages of Flask is its easy-to-understand and intuitive API, which allows developers to quickly get started with building web applications. Its modular structure allows for flexibility in choosing components and libraries based on project requirements. Flask also supports various extensions and plugins, making it highly customizable and adaptable to different use cases. With Flask, we can effortlessly handle routing, URL mapping, and request handling. The framework provides an elegant way to define routes and associate them with appropriate functions, allowing for clean and organized code structure. Flask's templating engine further simplifies the process of generating dynamic HTML content, enabling the seamless integration of server-side logic with client-side presentation.
- **Numpy:** Python's Numpy library ([46]) is a fundamental tool for scientific computing, offering high-level mathematical abstractions and efficient numerical computation capabilities. Numpy provides a powerful n-dimensional array object and serves as the primary

library for scientific computing in Python. It seamlessly integrates with other programming languages, such as C and C++, and offers a wide range of functionalities. The n-dimensional array type introduced by Numpy enhances the Python programming language with a versatile data structure that excels in performing numerical computations. This array object enables efficient storage and manipulation of large datasets, facilitating various scientific and mathematical operations. Additionally, Numpy includes essential numerical algorithms, including those for linear algebra and eigenvector identification, which greatly simplify complex calculations. Moreover, Numpy plays a significant role in scientific applications by providing features such as random number generation and other tools. These functionalities are invaluable in fields such as data analysis, simulation, and machine learning, where randomization and statistical analysis are essential.

- **Scikit Learn:**

Scikit-learn, a popular and free machine learning library ([45]) for Python, offers a comprehensive range of functionalities for data analysis and modeling. With seamless integration with other Python numerical and scientific libraries like NumPy and SciPy, Scikit-learn provides a versatile and powerful toolkit for classification, regression, and clustering tasks. It incorporates a diverse set of algorithms, including support vector machines, random forests, and k -nearest neighbors. Python serves as the primary programming language for Scikit-learn, ensuring ease of use and accessibility for developers. Moreover, Scikit-learn is designed for optimal efficiency, with several essential algorithms implemented directly in Python. For instance, support vector machines are supported through a Python wrapper built around LIBSVM, while logistic regression and linear support vector machines utilize a similar wrapper over LIBLINEAR.

- **Tensorflow:**

TensorFlow, an open-source and freely available software library [43], is widely used for numerical computations and data flow graph-based calculations. It utilizes a graph-based representation, where nodes in the graph represent mathematical formulas, and edges represent multidimensional arrays known as tensors that are transferred between the nodes. With a unified API, TensorFlow enables computation deployment across CPUs, GPUs, and other devices in desktops, servers, and mobile devices. TensorFlow's extensive ecosystem provides a vast collection of pre-built models, tools, and resources, making it easier for developers to leverage state-of-the-art techniques in their projects. It supports distributed computing and efficient execution across multiple devices, enabling the scaling of

models to handle large datasets and complex computations. By combining computational power with ease of use and flexibility, TensorFlow has become one of the most popular libraries for machine learning and numerical computation. Its wide adoption and active community contribute to its continuous development and advancements in the field of artificial intelligence and scientific computing.

- **Keras:**

Keras, a high-level neural networks API, is a Python library ([47]) that can be run on top of CNTK, TensorFlow, or Theano. It is specifically designed to facilitate rapid experimentation during development, aiming to minimize the time from concept to results. Keras prioritizes user-friendliness, modularity, and extensibility, enabling researchers to quickly create prototypes. One of the key advantages of Keras is its compatibility with both CPU and GPU, allowing for efficient computation on various hardware configurations. It supports convolutional and recurrent networks, as well as combinations of the two, providing flexibility for a wide range of neural network architectures. Keras offers a comprehensive set of tools for building neural networks, including pre-built implementations of commonly used elements such as layers, objectives, activation functions, and optimizers. Additionally, the library provides convenient utilities for handling image and text data, simplifying data preprocessing and integration into neural network models. The Keras source code is hosted on GitHub, providing transparency and easy access for developers. The community support for Keras is robust, with communication channels including a Glitter channel, a Slack channel, and a dedicated problems page on GitHub. These resources enable developers to seek assistance, share knowledge, and collaborate with other Keras users.

- **OpenCV:**

OpenCV (Open Source Computer Vision) is a powerful programming library ([44]) primarily focused on real-time computer vision applications. Its adoption is driven by the simplicity it offers in performing a wide range of tasks without the need for extensive implementation. With Python as its base language, OpenCV provides a modular neural network library that can be integrated with TensorFlow. OpenCV is designed to facilitate quick experimentation, enabling users to swiftly translate their ideas into tangible outcomes. The choice of TensorFlow as a backend was motivated by its deep learning capabilities, as both TensorFlow and Keras were purpose-built to address the challenges associated with deep learning. Being implemented in Python, both systems offer a concise

and streamlined user experience, eliminating the need for managing multiple files simultaneously. One of the notable advantages of both TensorFlow and Keras is their ability to efficiently leverage both CPU and GPU resources, resulting in high-speed computations. This allows for accelerated processing and training of deep learning models, enhancing overall performance and productivity.

4.2 Implementation

For comparing and evaluating different deep learning models used in this thesis, certain metrics are used.

4.2.1 Experimental Evaluations (Metrics)

In order to evaluate how well different models work, we make use of the following metrics:

1. **Confusion Matrix**: The confusion matrix helps us understand how well our classifier is performing by revealing the types of errors made. It allows us to analyze the true positives, true negatives, false positives, and false negatives. These elements in the confusion matrix are defined as follows:
 - True Positives (TP): The number of instances correctly predicted as a specific class.
 - True Negatives (TN): The number of instances correctly predicted as not belonging to a specific class.
 - False Positives (FP): The number of instances incorrectly predicted as belonging to a specific class when they do not.
 - False Negatives (FN): The number of instances incorrectly predicted as not belonging to a specific class when they do.
2. **Accuracy**: Accuracy provides a measure of how well the model performs overall, indicating the proportion of correct predictions in relation to the total number of predictions. It measures the overall correctness of the model's predictions by comparing the number of correct predictions to the total number of predictions made.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)} \quad (4.1)$$

3. **Precision**: Precision provides insights into the model's ability to avoid false positives. It quantifies the model's accuracy when predicting positive instances, indicating the propor-

tion of correctly predicted positives out of all instances predicted as positive. A higher precision value indicates a lower rate of false positives, suggesting a more reliable positive prediction from the model.

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \quad (4.2)$$

4. **Recall**: Recall provides insights into the model's ability to correctly identify positive instances. It quantifies the model's ability to avoid false negatives, indicating the proportion of correctly predicted positives out of all actual positive instances. A higher recall value suggests that the model is effective in capturing positive instances and has a lower rate of false negatives.

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \quad (4.3)$$

5. **F1-score**: It provides a balance between precision and recall, considering both false positives and false negatives.

$$\text{F1 score} = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN}) \quad (4.4)$$

By calculating the harmonic mean of precision and recall, the F1 score provides a single metric that captures the trade-off between precision and recall. It balances the ability of the model to avoid false positives and false negatives. Its range lies from 0 to 1, where a higher value indicates better performance. A perfect F1 score of 1 indicates a model with both perfect precision and recall, while a score of 0 indicates poor performance in terms of precision and recall.

4.2.2 Hyperparameter

- **Optimizer**: The Adam optimizer is a popular optimization algorithm used in training deep neural networks. It combines the advantages of AdaGrad and RMSProp techniques to offer efficient and adaptive updates to the model's parameters. With Adam, each parameter has its own adaptive learning rate, which helps address issues with sparse gradients and facilitates faster convergence. It achieves this by calculating different adaptive learning rates for each parameter based on the history of gradients and squared gradients. The adaptive learning rates in Adam are computed as a weighted average of past gradients, similar to RMSProp, and a momentum term inspired by the concept of moving averages. These adaptive learning rates enable Adam to handle different magnitudes of gradients

for each parameter, resulting in effective updates and improved convergence speed.

- **Loss:** In multi-class classification, each data instance is assigned to one of several possible classes. Categorical Cross-Entropy calculates the dissimilarity between the predicted probabilities for each class and the actual target labels. It compares the predicted probabilities with the true class labels to determine how well the model is performing. The Categorical Cross-Entropy loss function takes into account the entire probability distribution produced by the model, rather than just the predicted class. It encourages the model to assign higher probabilities to the correct class and lower probabilities to incorrect classes. By minimizing the Categorical Cross-Entropy loss during training, the model learns to make more accurate predictions and assigns higher probabilities to the correct classes. This loss function is a valuable tool in training multi-class classification models and helps improve their performance by reducing the discrepancy between predicted and true class probabilities.
- **Metrics:** This parameter is not required, but it does give the user the ability to check the accuracy of the model after each stage of training. After the model has been compiled, it is now possible to begin the training. "fit" is the name of the Keras function that is used to train a model, and it has a large number of parameters that can also be adjusted. Some of these parameters are as follows:
 - Train Data: It is an array of the training data stored in Numpy.
 - Train Test: It is an array of the target data represented in Numpy.
 - Batch Size: It is the number of samples that are updated with each gradient.
 - Number of epochs: It is the number of times that the training data should be iterated over.
 - Callbacks: With the help of this option, the user is given the opportunity to save the weights of the network after each epoch, provided that the loss is less than any value that came before it.
 - Validation Data: This refers to the sub-set of data that is used to validate how well the model is working.
 - Shuffle: It gives the user the ability to automatically mix up the training data after each iteration of the algorithm.

ID	Literature	Year	CNN models	Validation accuracy	Testing accuracy
1	Tm et al.[38]	2018	LeNet	94.95%	NA
2	Sardogan et al.[32]	2018	LVQ CNN	86.00%	NA
3	Hari et al.[33]	2019	PDDNN	86.00%	NA
4	Geetharamani et[34]	2019	Nine-layer DCNN	96.46%	NA
5	Karthik et al. [39]	2019	Attention-based residual CNN	98.00%	NA
6	Chen et al.[42]	2020	VGGNet Inception	91.83%	NA
7	Argueso et al.[36]	2020	Inception V3	91.40%	NA
8	Tuncer et al.[37]	2021	Hybrid CNN	99.00%	NA
9	Wang et al.[40]	2021	Attention-based NN with Bayesian optimization	NA	94.65%
10	Sowmyalakshi et al.[41]	2021	CNNIR-OWELM-based deep learning	NA	94.20%
11	Latif et al.[1]	2022	MobInc-Net	99.21%	95.62%
12	This study	2023	Multihead Attention-based VGG model	98.89%	97.2%

Table 4.1: Comparison of Validation and Testing Accuracy on Different Models: Comparative Analysis with Related Research Papers

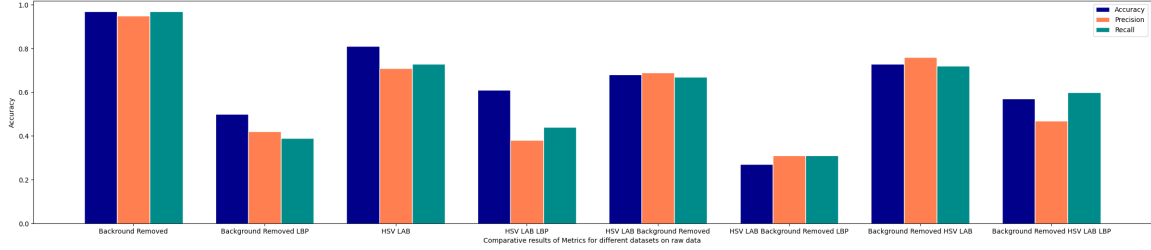


Figure 4.1: Average comparison of metrics when the trained model was tested on raw images

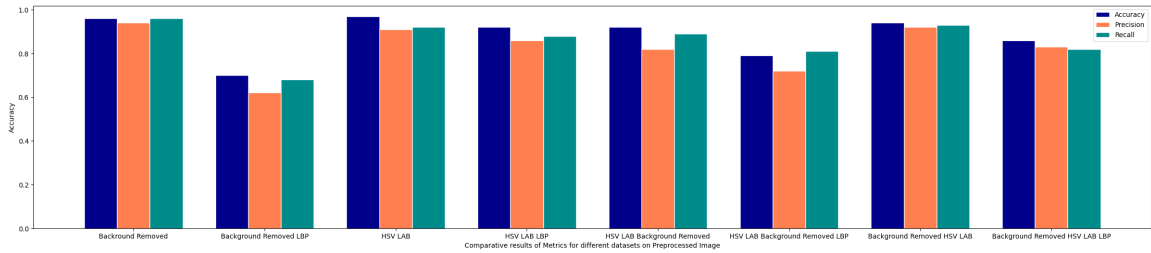


Figure 4.2: Average comparison of metrics when the trained model was tested on processed images

4.2.3 Experimental Setup and Model Assesment

Our dataset consisted of 44,569 images of paddy crops, including healthy crops and those affected by five types of diseases: Bacterial Leaf Blight, Brown Spot, Leaf Blast, False Smut,

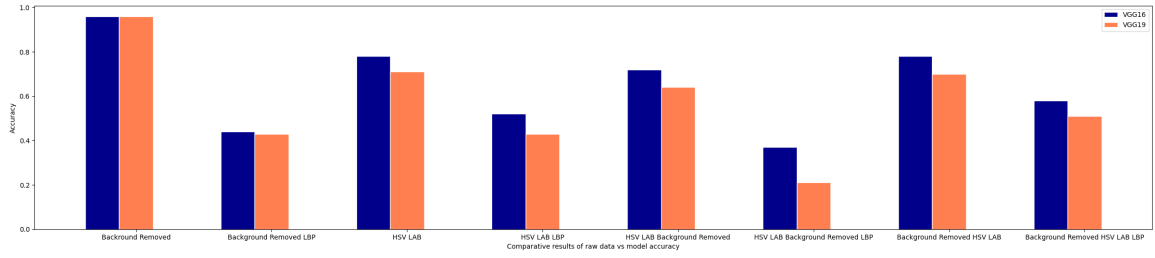


Figure 4.3: Average comparison of accuracy when the trained model was tested on raw images

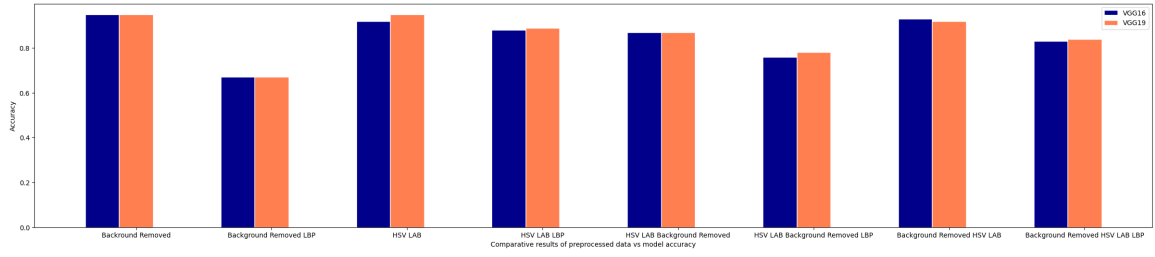


Figure 4.4: Average comparison of VGG16 and VGG19 model's accuracy on processed data

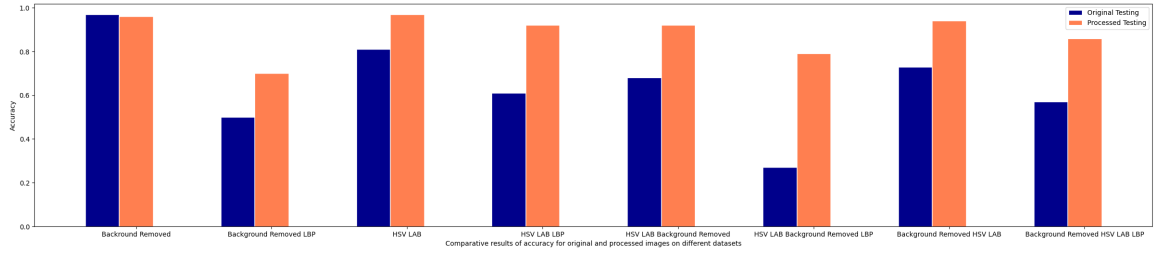


Figure 4.5: Average comparison of accuracy for original and processed data

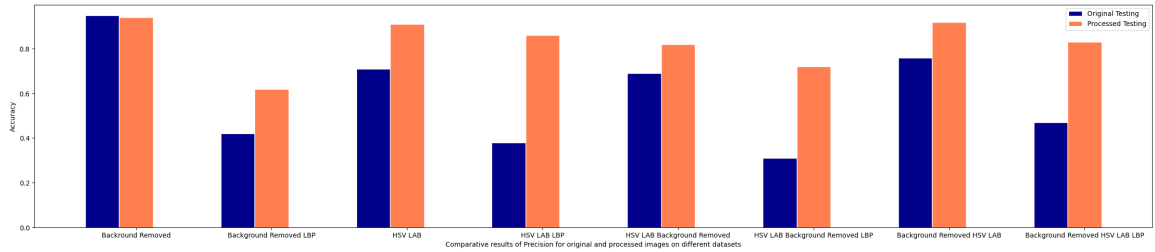


Figure 4.6: Average comparison of precision for original and processed data

and Sheath Blight. To handle the large size of the dataset, we used a Convolutional Neural Network (CNN) classifier, which is known to perform well on large datasets, and has a relatively shorter testing time compared to other classifiers such as Support Vector Machines (SVM). Additionally, CNNs work very well with high-dimensional data like images. To further improve the performance of our model, we employed transfer learning, a technique in which a pre-trained model is used as a starting point instead of training from scratch. We used the VGG-16 and

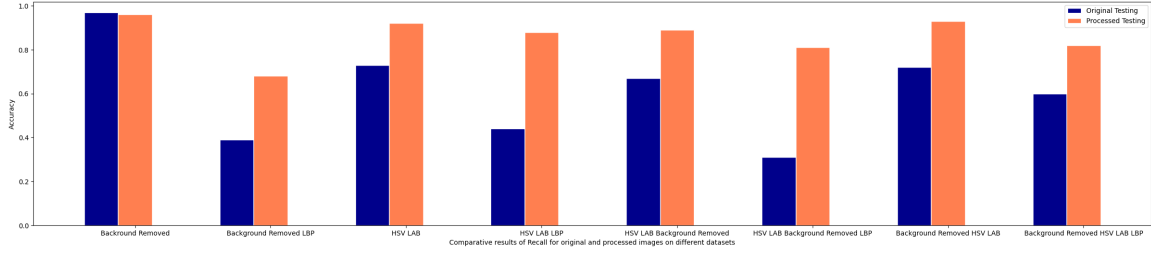


Figure 4.7: Average comparison of recall for original and processed data

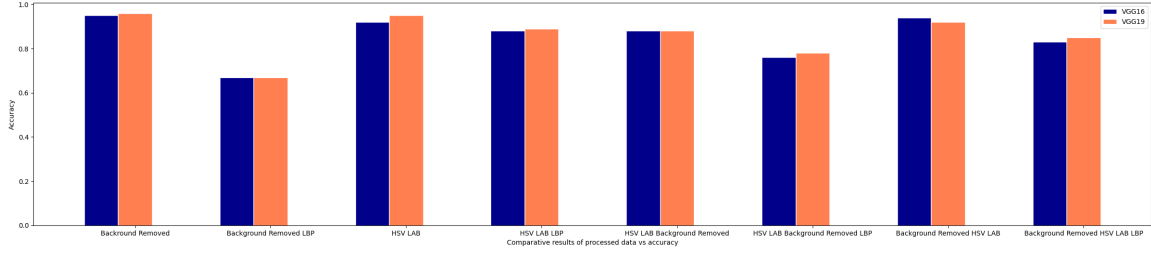


Figure 4.8: Average comparison of VGG16 and VGG19 model on the basis of accuracy

VGG-19 pre-trained models as they have a depth of 16 and 19 layers, respectively, which have shown to perform well on a variety of image classification tasks. We chose VGG models over other pre-trained models like InceptionV3, ResNet50, and MobileNet, which have 189, 107, and 55 layers, respectively, due to their simplicity and ease of modification and adaptation for different tasks.

The dataset was split into training, testing, and validation sets, with 70% used for training, 20% for testing, and 10% for validation. We used eight different pre-processed datasets for training the models: Background Removed, Background Removed LBP, HSV LAB, HSV LAB LBP, Background Removed HSV LAB, Background Removed HSV LAB LBP, HSV LAB Background Removed, and HSV LAB Background Removed LBP. We used three metrics to evaluate the models: accuracy, precision, and recall. For each pre-processed dataset, we trained VGG-16 and VGG-19 models, resulting in a total of 48 models trained. During the training process, we used weights from ImageNet to initialize the model and transfer the learned features from a wide range of images. Since pre-trained models have already learned to detect generic features, they are effective in image recognition tasks and provide state-of-the-art performance. Finally, we saved the trained models based on the best validation score for each metric. The trained models can be used to predict the type of disease affecting paddy crops, which can aid farmers in taking timely measures to prevent crop damage. During the Model Assessment phase, the best-performing models were evaluated on the test set of the processed dataset. The comparison between the original and predicted labels of the test images was performed to ob-

Dataset	Model	Original Testing			Processed Testing		
		Accuracy	Precision	Recall	Accuracy	Precision	Recall
Background Removed	VGG16	0.96429	0.93731	0.97145	0.95611	0.93403	0.95934
	VGG19	0.96915	0.95970	0.96022	0.96164	0.94551	0.95866
HSV LAB	VGG16	0.77910	0.67654	0.68268	0.96712	0.91456	0.87646
	VGG19	0.84953	0.73399	0.77830	0.97187	0.91279	0.96148
Background Removed + LBP	VGG16	0.53956	0.40261	0.35861	0.69582	0.63716	0.67171
	VGG19	0.46258	0.44718	0.41712	0.69755	0.61091	0.69823
HSV LAB + LB	VGG16	0.69823	0.22249	0.42046	0.91957	0.84232	0.89138
	VGG19	0.55786	0.54478	0.46673	0.91467	0.88601	0.86440
Background Removed + HSV LAB	VGG16	0.65802	0.73176	0.69988	0.94885	0.91827	0.94029
	VGG19	0.81146	0.78241	0.73687	0.93481	0.91712	0.91717
Background Removed + HSV LAB + LBP	VGG16	0.57448	0.36868	0.59541	0.84828	0.82056	0.81498
	VGG19	0.57448	0.57599	0.61075	0.87782	0.83257	0.82667
HSV LAB + Background Removed	VGG16	0.66112	0.62850	0.63152	0.91419	0.80604	0.90892
	VGG19	0.69958	0.74603	0.71660	0.91623	0.83162	0.87870
HSV LAB + Background Removed + LBP	VGG16	0.20626	0.26164	0.17380	0.79393	0.70980	0.79388
	VGG19	0.32928	0.35715	0.44108	0.79075	0.72859	0.81774

Table 4.2: Comparing Results of Models on Different Datasets: Performance Evaluation across Various Experimental Settings

tain the percentage of correctly predicted samples. This evaluation process yielded very good results for most of the saved models. All the entries for the original dataset were shuffled along with their corresponding labels to prevent the models from being biased toward certain classes. Finally, testing was carried out on two sets of test images, including the original images without pre-processing and the images in the original dataset that were pre-processed.

Now, coming to another multi-model approach for early prediction of harvesting cost of the field in this we used multi-head attention features combined with machine learning algorithms to extract useful features for predicting paddy crop harvesting costs. The proposed system employed data cleaning and feature selection stages, eliminating inconsistencies and missing values while identifying pertinent features using the mutual info regressor. Regression models, including linear regression, ridge regression, lasso regression, decision tree regression, K-nearest neighbors (KNN), and XGBoost, were then employed for prediction. The performance of each model was evaluated using k-fold cross-validation, and the R-squared (R²) score was used as the evaluation metric to assess the accuracy of the predictions. This comprehensive approach

ensured a robust and accurate analysis of the paddy crop dataset, providing reliable estimates for harvesting costs.

Further, for land fertilization prediction, we utilized multi-head attention to integrate information on soil quality, weather patterns, and crop rotation, enabling optimal recommendations for farmers regarding fertilizer type and quantity. The system considered various factors and provided timely and accurate information to facilitate informed decision-making, leading to improved crop yields and cost savings. In this module, our objective was to predict the values of Nitrogen (N), Phosphorus (P), and Potassium (K) components of fertilizers based on input parameters such as crop types and crop location. We split the fertilizer dataset into training and testing datasets, with 80% of the data allocated for training the model and 20% for testing. Three separate random forests (RF) classifiers, each comprising 50 decision trees, were employed to predict N, P, and K values. The accuracy of the overall predictions was measured by averaging the accuracy of the individual decision trees. The dataset, consisting of $n = 2200$ entries, was divided into training and test datasets, with 1760 entries for training and 240 entries for testing. The random forest regression algorithm was applied, utilizing 50 estimators or decision trees, to predict the values of N, P, and K. Each value had a specific model (N Label, P Label, K Label) trained using the training dataset, with the dependent variable set accordingly. The models generated outputs consisting of 50 decision trees, collectively forming the random forest regression models for each label. This training process ensured the models were capable of predicting Nitrogen, Phosphorus, and Potassium values based on the provided dataset.

4.2.4 Model Performance Evaluation

Based on the performance of various models across processed datasets and metrics, we adopted seven models with the highest percentage of correct results. These include VGG16 ACCURACY Background Removed HSV LAB, VGG16 RECALL Background Removed HSV LAB, VGG19 ACCURACY Background Removed, VGG19 RECALL Background Removed, VGG19 ACCURACY HSV LAB, VGG16 ACCURACY HSV LAB, and VGG19 RECALL HSV LAB.

We found that processed datasets like HSV LAB, Background Removed, and Background Removed HSV LAB generated the best results. The HSV LAB dataset was useful in segmenting the diseased from the non-diseased portion of the image, while the Background Removed dataset was useful in segmenting the background noise from the rest of the image. The Background Removed HSV LAB dataset was particularly effective in segmenting both the background noise and non-diseased portion of the image from diseased patterns, respectively.

Regressor	Training R2-Score	Testing R2-Score
Linear Regressor	95%	91%
Decision Tree	98%	93%
Lasso	95%	91%
Ridge	95%	91%
KNN	90%	87%
XG-Boost	98%	94%

Table 4.3: Comparison of Training and Testing R2-Score Results on Various Models: Evaluating Performance Across Different Algorithms

However, we also found that the LBP version of the datasets was not very useful since they could not distinguish between the essential and non-essential portions of the image and were completely reliant upon the previous processing adopted. Additionally, in the case of HSV LAB Background Removed, a substantial portion of the diseased portion of the image was also removed, leading to the loss of useful features and necessary information.

Interestingly, we found that testing results on processed images were superior compared to testing results on original images. However, in the case of the Background Removed dataset, testing results on original images were better than processed images, as shown in figure(4.1).

When evaluating the performance of various models on different datasets, we observed that the metrics of accuracy and recall were consistently better than precision for testing on processed images. This trend was evident across all datasets, as indicated by the selected models that exhibited higher accuracy and recall metrics, figure(4.2) shows the average comparison of metrics when the trained model was tested on processed images.

However, there was no clear pattern for metric comparison across the datasets for testing on original images, figure(4.4) shows the average comparison of the VGG16 and VGG19 model's accuracy on processed data. Additionally, the performance of VGG19 was consistently superior to VGG16 across all datasets when testing on original images, figure(4.3) shows the average comparison of accuracy when the trained model was tested on raw images. Figure (4.9,4.10,4.11,4.12) shows the comparison of model accuracy and model loss comparison for VGG16 and VGG19 models.

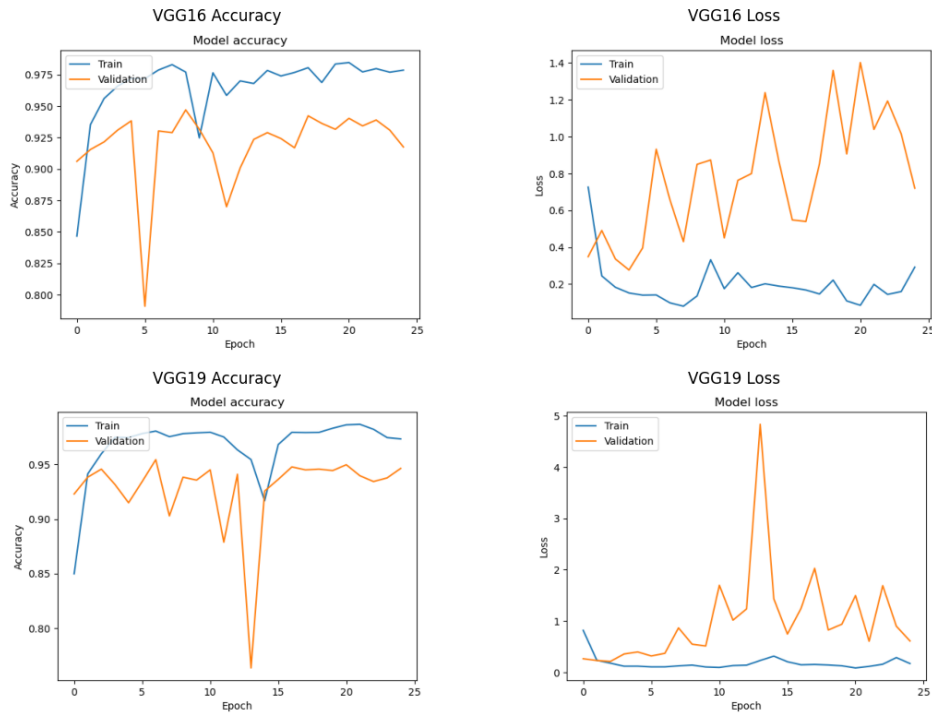


Figure 4.9: Performance comparison of VGG16 and VGG19 models on Background Removed + HSV Lab dataset

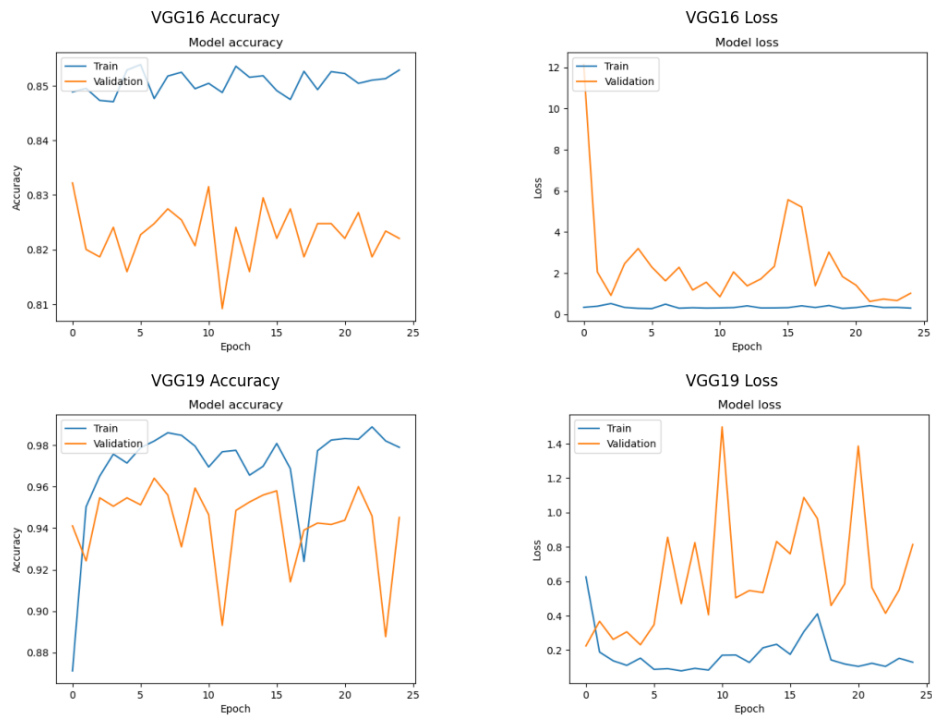


Figure 4.10: Performance comparison of VGG16 and VGG19 models on HSV lab Background Removed dataset

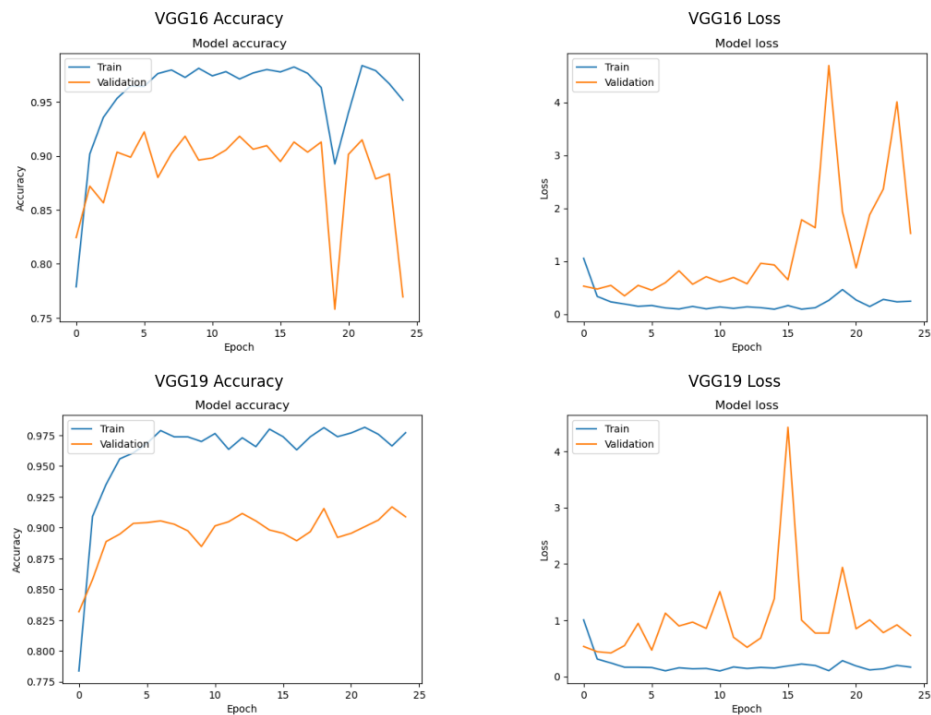


Figure 4.11: Performance comparison of VGG16 and VGG19 models on HSV lab Background Removed dataset



Figure 4.12: Performance comparison of VGG16 and VGG19 models on HSV lab Background Removed LBP dataset

Conclusion and Future Work

The research study presented substantial contributions in three crucial areas: rice disease detection, harvesting cost prediction, and land fertilization prediction. In the realm of rice disease detection, the study proposed and implemented an effective strategy that leveraged image pre-processing techniques, including background removal and segmentation, combined with multi-head attention models.

Moving on to harvesting cost prediction, the study introduced a comprehensive methodology that combined multi-head attention features with machine learning algorithms. By considering multiple factors such as farm size, seed varieties, fertilizer usage, labor hours, and more, the proposed system enabled more accurate estimation of harvesting costs. Utilizing multi-input and multi-output models, it provided farmers with precise cost predictions, empowering them to optimize their operations, allocate resources effectively, and enhance financial outcomes.

In the domain of land fertilization prediction, the research study addressed the critical aspect of optimizing fertilizer usage for improved crop yield and sustainability. By integrating information on soil quality, weather patterns, crop rotation, and other relevant factors through multi-head attention models, the proposed system offered farmers optimal recommendations regarding the type and quantity of fertilizers to be used. This multi-input and multi-output approach not only maximized crop productivity but also minimized environmental impact by reducing excess fertilizer application, promoting sustainable farming practices. The research outcomes lay the foundation for further exploration and refinement of these techniques, paving the way for future advancements in agricultural technology and improved outcomes for farmers worldwide.

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Publications

Journal Proceedings:

- **Vinay Mishra**, N. Srinivas Naik and Santosh Kumar “MultiModal Crop Decision Support System for Paddy Cultivation using Multi-Head Attention-Based Multi-Input Multi-Output Model ”, in **Jornal of Computers and Electronics in Agriculture**, Under Review, submitted on 13th June 2023.
- Shivanshi Gupta, **Vinay Mishra** and Avantika Singh “Computerized Chest X-ray Report Generation via Masked Multi-Head Attention & Pathology Tags, in **Journal of Multimedia System (SCI Journal)**, Under Review, submitted on 30th May 2023.

Conference Proceedings:

- **Vinay Mishra**, N. Srinivas Naik and Santosh Kumar “Maize Plant Disease Prediction on UAV images using Four Stage Ensemble Pipeline”, in **ICCIT, 2023, IEEE, Saudi Arabia Section**, Accepted.



Author's Biography

Vinay Mishra, born in the city of Raipur, Chhattisgarh, is an accomplished individual with a passion for learning and a drive for innovation. He completed his schooling at Krishna Public School in Raipur, where he developed a strong academic foundation and nurtured his thirst for knowledge. Inspired by the transformative potential of technology, Vinay pursued a Bachelor of Technology (B.Tech) degree from the renowned Bhilai Institute of Technology, Durg. During his undergraduate studies, he discovered the immense power of technology in addressing real-world challenges and found his calling in the field.

Motivated to further expand his expertise, Vinay embarked on a journey to pursue a Master of Technology (M.Tech) degree from IIIT Naya Raipur, specializing in Data Science and Artificial Intelligence. This advanced program provided him with a deep understanding of data analysis, machine learning, and AI algorithms.

In his research pursuits, Vinay focused on exploring the potential of AI in revolutionizing agriculture, seeking to optimize farming practices, improve crop yields, and promote sustainable agricultural solutions. Additionally, he delved into the field of medical image diagnosis, harnessing the power of AI algorithms to enhance the accuracy and efficiency of disease detection. Driven by a strong commitment to make a positive impact, Vinay strives to contribute to the ever-evolving field of AI through innovative research and practical applications in critical sectors such as agriculture and healthcare.



